

Space: Retail-IP (RETAILIP) · Author: Anjali Jaiswal

Source: <https://flipkart.atlassian.net/wiki/spaces/RETAILIP/pages/491390643/Meridian+Phase+1+UI+API+Contract> (+ all subpages)

Meridian Phase 1 — UI-API Contract (Overview)

Base URL: `/api/v1` | **Auth:** Bearer token (corporate SSO) | **Errors:** RFC 7807 problem+json | **Timestamps:** ISO 8601, `Asia/Kolkata`

All curl examples use `http://{instockctl-host}` as the service host. Every request requires `--header 'Authorization: Bearer {sso-token}'`.
POST requests also require `--header 'Content-Type: application/json'`.

Complete request/response specification for all API endpoints powering Dashboard A (Replenishment Execution), Dashboard B (Instock Monitoring), and Dashboard C (Instock Diagnostics). The contract is split into subpages by domain — each subpage is included in full below, in order.

Subpages

#	Section	Endpoints
0	Error Responses	Common RFC 7807 error shapes
1	Generic	Filters, Status, Exports (GET + POST + job status)
2	Instock Monitoring — KPI Cards	KPI headline metrics, worst FSNs, low-instock stores
3	Instock Monitoring — City Instock Table	City table, sub-pivots, dark stores, product drilldown, 4 RCA modals, async export
4	Instock Diagnostics	FSN×DS diagnostic, FSN×SH diagnostic, consolidated bulk export, dynamic pivot
5	Replenishment Execution	Live 15-day grid, cell drill-down, STN list, consignment detail
6	Cron Monitoring — Summarised View	City, SH, and Network Component allocation summaries
7	Cron Monitoring — Pivoted Deep Dive	Trend view, run level view, runs drill-in, run constraints
8	Capacity Utilisation	SH capacity, DS capacity, lane view, cell drill-down, under-utilisation, wasted capacity
9	Bulk Actions	Bulk approve / reject / override requirements
—	Navigation Flows & Frontend Cheat Sheet	Every click-chain mapped to its API call sequence

Error Responses

All endpoints return RFC 7807 `application/problem+json` on error.

HTTP Status	type	When
400 Bad Request	<code>urn:meridian:error:bad-request</code>	Missing required parameter, invalid enum value, date range > 14 days.
401 Unauthorized	<code>urn:meridian:error:unauthorized</code>	SSO token absent, expired, or invalid.
403 Forbidden	<code>urn:meridian:error:forbidden</code>	Authenticated but not authorised for the resource.

HTTP Status	type	When
404 Not Found	urn:meridian:error:not-found	jobId , stnId , or runId does not exist.
422 Unprocessable	urn:meridian:error:unprocessable	Logical constraint violated (e.g. scope=CITY without cityCode).
500 Internal Server Error	urn:meridian:error:internal	Unexpected server error.

Error body example (400):

```
{
  "type": "urn:meridian:error:bad-request",
  "title": "Bad Request",
  "status": 400,
  "detail": "Parameter 'scope' is required.",
  "instance": "/api/v1/instock/rca/supply-chain"
}
```

1. Generic — Filters, Status & Exports

Shared APIs used across all dashboard pages: filter options, pipeline status, and the async export queue / download manager.

GET /api/v1/filters/options — Filter List Options

Returns all dimension values for dropdowns across pages — cities, sourcing hubs, dark stores, FSNs, FSN categories, FSN portfolios, BUs, super categories, verticals, brands, network components, store age bands, and store OPD bands. Pass a subset via `sections` to reduce payload.

Parameters

Parameter	In	Type	Required	Description	Example
<code>sections</code>	query	string[] (CSV)	No	Subset of filter groups to return. Values: <code>cities</code> <code>sourcingHubs</code> <code>darkStores</code> <code>fsns</code> <code>fsnCategories</code> <code>fsnPortfolios</code> <code>bus</code> <code>superCategories</code> <code>verticals</code> <code>brands</code> <code>networkComponents</code> <code>storeAgeBands</code> <code>storeOpdBands</code> . Default: all groups.	<code>cities,darkStores</code>
<code>city</code>	query	string[] (CSV)	No	When provided alongside <code>sections=darkStores</code> or <code>sections=sourcingHubs</code> , narrows returned options to the given city codes.	<code>BLR,DEL</code>

Response Schema

Field	Type	Description
<code>cities</code>	Array<{code, name}>	All active city codes and display names.
<code>sourcingHubs</code>	Array<{code, name, city}>	All SH codes with display name and parent city.
<code>darkStores</code>	Array<{code, name, city, primarySH}>	All DS / FC codes with display name, city, and primary sourcing hub.
<code>fsns</code>	Array<{fsn, productName}>	All active FSN identifiers with human-readable product names.
<code>fsnCategories</code>	string[]	Fixed set: T100, T500, T2500, T5000, T7500, Rest, Festive, Sampling.
<code>fsnPortfolios</code>	string[]	Fixed set: Dry, DBEFM.
<code>bus</code>	string[]	All Business Unit names.
<code>superCategories</code>	string[]	All Super Category names.
<code>verticals</code>	string[]	All Vertical names.
<code>brands</code>	string[]	All Brand names.

Field	Type	Description
networkComponents	Array<{id, label, city}>	Network component composite keys grouped by city.
storeAgeBands	string[]	Fixed bands: 0–15 , 15–30 , 30–60 , >60 (days since store launch). Confirmed 2026-07 — no "days" suffix in the actual values ; matches the GQL schema convention. The word "days" is unit context only, not part of the string.
storeOpdBands	string[]	Fixed bands: <500, 500-1000, 1000-1500, 1500-2000, 2000-2500, 2500-3000, >3000.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/filters/options?sections=cities,darkStores&city=BLR' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "cities": [
    { "code": "BLR", "name": "Bengaluru" },
    { "code": "DEL", "name": "Delhi" }
  ],
  "darkStores": [
    { "code": "BLR_DS_001", "name": "Koramangala DS", "city": "BLR", "primarySH": "BLR_SH_01" },
    { "code": "BLR_DS_002", "name": "Indiranagar DS", "city": "BLR", "primarySH": "BLR_SH_01" }
  ]
}
```

GET /api/v1/status — Pipeline & Data Freshness Status

Returns the latest pipeline run metadata. Used by the frontend to render "Data as of HH:MM" on every dashboard page, show a hold banner when the anomaly gate fires, and enable ops monitoring. Should be polled on page load and can be cached for up to 5 minutes client-side.

Parameters

None.

Response Schema

Field	Type	Description
lastRefreshId	string	Refresh ID of the most recently published pipeline run.
lastRunTimestamp	ISO 8601	Timestamp when the last successful pipeline run completed. Drives "Data as of HH:MM" display.
pipelineCadenceHrs	number	Configured cadence in hours (default 4).
nextExpectedRun	ISO 8601	Approximate timestamp of the next scheduled run.
isHeld	boolean	True when the anomaly gate (J6 exit code 43) fired and the last run was held.
holdReason	string null	Human-readable reason the run was held. Null when <code>isHeld=false</code> .

Field	Type	Description
holdSince	ISO 8601 null	Timestamp when the hold began. Null when <code>isHeld=false</code> .

Curl

```
curl --location 'http://{instockctl-host}/api/v1/status' \  
--header 'Authorization: Bearer {sso-token}' \  
--header 'Accept: application/json'
```

Example Response

```
{  
  "lastRefreshId": "20260622_1400",  
  "lastRunTimestamp": "2026-06-22T14:32:00+05:30",  
  "pipelineCadenceHrs": 4,  
  "nextExpectedRun": "2026-06-22T18:32:00+05:30",  
  "isHeld": false,  
  "holdReason": null,  
  "holdSince": null  
}
```

GET /api/v1/exports — Download / Action History ("Downloads Manager")

Returns all async export jobs initiated by the authenticated user within the last 24 hours, newest first. Used by the Download Manager panel accessible from any dashboard header.

Reconciled 2026-07 against the Figma "Downloads Manager" read (from the Instock Diagnostics Consolidated Report flow, page 4) — confirms the shape below but raises two discrepancies, flagged not resolved:

- **Confirmed:** `scopeLabel` 's intended content — Figma examples ("Mumbai | Supply Chain Delay | TL Not Created | FSN00010018 | DS_005", "2 FSN x 3 DS") match a human-readable breadcrumb-style label, consistent with this field as already documented.
- **Open — job ID display format:** Figma shows a request ID like `EXP-MR34U5BI` in the UI, not a raw UUID. Unconfirmed whether that's a separate human-readable display ID alongside the internal `jobId` (UUID), or whether `jobId` itself should switch format. Don't assume either — confirm with backend before the UI hardcodes a parser for one format or the other.
- **Open — status enum simplification:** Figma's Downloads Manager only shows 3 badge states — `Completed` / `Failed` / `In-progress` — vs. the 5-value enum below (`QUEUED` / `PROCESSING` / `READY` / `FAILED` / `EXPIRED`). This could just be a UI-side collapse (`QUEUED+PROCESSING` → "In-progress" badge, `READY` → "Completed") rather than a backend enum change — flagging as evidence for the open question already noted on `GET /api/v1/exports/{jobId}` below, not a resolution of it.
- **Open — entry point:** where "Downloads Manager" is navigated to from wasn't visible in the Instock Diagnostics screenshots read so far (not in the left icon rail). Unconfirmed whether it's a top-bar icon, a post-submit link, or something else.

Parameters

Parameter	In	Type	Required	Description	Example
status	query	enum	No	Filter by job state. Values: <code>QUEUED</code> <code>PROCESSING</code> <code>READY</code> <code>FAILED</code> <code>EXPIRED</code> .	<code>READY</code>
limit	query	integer	No	Max records. Default 50.	<code>20</code>
offset	query	integer	No	Pagination offset. Default 0.	<code>0</code>

Response Schema

Field	Type	Description
jobs	Array<ExportJob>	Ordered by requestedAt descending.
jobs[].jobId	string (UUID)	Unique job identifier. See open question above re: display format vs. Figma's <code>EXP-XXXXXXX</code> id.
jobs[].scopeLabel	string	Human-readable export scope, e.g. "All Cities — Electronics". Confirmed against Figma examples — see note above.
jobs[].exportType	string	<code>OOS_FULL</code> <code>SC_RCA</code> <code>CRON_RCA</code> <code>NORM_RCA</code> <code>STN_RCA</code> <code>REPLENISHMENT_EXECUTION</code> <code>DIAGNOSTICS_CONSOLIDATED</code> .
jobs[].status	enum	Current job state.
jobs[].estimatedRows	integer	Row count estimate.
jobs[].requestedAt	ISO 8601	Submission timestamp.
jobs[].completedAt	ISO 8601 null	Completion timestamp. Null until done.
jobs[].downloadUrl	pre-signed URL null	Non-null only when <code>status=READY</code> . Expires 24h after generation.
jobs[].errorReason	string null	Human-readable error message when <code>status=FAILED</code> . Matches the Figma "hover info icon for failure reason" behaviour.
total	integer	Total matching jobs for pagination.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/exports?limit=3' \
  --header 'Authorization: Bearer {sso-token}' \
  --header 'Accept: application/json'
```

Example Response

```
{
  "jobs": [
    {
      "jobId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
      "scopeLabel": "All Cities — Electronics",
      "exportType": "OOS_FULL",
      "status": "READY",
      "estimatedRows": 42800,
      "requestedAt": "2026-06-22T10:15:00+05:30",
      "completedAt": "2026-06-22T10:17:43+05:30",
      "downloadUrl": "https://storage.googleapis.com/meridian-exports/a1b2c3d4...?token=...",
      "errorReason": null
    }
  ],
  "total": 1
}
```

POST /api/v1/exports — Trigger Async Export

Enqueues a new async export job. Returns immediately with a `jobId` . Poll `GET /api/v1/exports/{jobId}` for status. Clients auto-refresh the Download Manager every 15 s while any job is `QUEUED` or `PROCESSING` .

The Instock Diagnostics "Consolidated Report Download" flow (page 4) has its own dedicated trigger endpoint (`POST /api/v1/diagnostics/consolidated-report`) rather than calling this generic one with `exportType=DIAGNOSTICS_CONSOLIDATED` . The dedicated `POST /api/v1/instock/export` on the Instock Monitoring page (page 3) follows the same intentional pattern — a page-specific trigger alongside this shared one, not a duplication to be consolidated. Confirmed 2026-07.

Parameters

Parameter	In	Type	Required	Description	Example
exportType	body	string	Yes	00S_FULL SC_RCA CRON_RCA NORM_RCA STN_RCA REPLENISHMENT_EXECUTION DIAGNOSTICS_CONSOLIDATED .	00S_FULL
scopeLabel	body	string	Yes	Human-readable label shown in Download Manager.	Bengaluru – Electronics
filters	body	FilterObject	Yes	Scope filters. Always includes optional arrays: <code>fsnCategory</code> , <code>fsnPortfolio</code> , <code>storeAge</code> , <code>storeOpd</code> . May include <code>cityCode</code> , <code>pivotType</code> , <code>pivotValue</code> , <code>darkStoreCode</code> , <code>sourcingHub</code> .	

Response Schema

Field	Type	Description
jobId	string (UUID)	Poll or download via <code>GET /api/v1/exports/{jobId}</code> .
status	QUEUED	Always QUEUED at submission.
estimatedRows	integer	Server-side row estimate.
estimatedSeconds	integer	Estimated generation time in seconds.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/exports' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Content-Type: application/json' \
--data-raw '{
  "exportType": "00S_FULL",
  "scopeLabel": "Bengaluru – All",
  "filters": {
    "cityCode": "BLR",
    "fsnCategory": ["T100", "T500"],
    "fsnPortfolio": [], "storeAge": [], "storeOpd": []
  }
}'
```

Example Response

```
{
  "jobId": "f9e8d7c6-b5a4-3210-fedc-ba9876543210",
  "status": "QUEUED",
  "estimatedRows": 18400,
```



```
"estimatedSeconds": 45
}
```

GET /api/v1/exports/{jobId} — Export Job Status

Returns current status, progress %, and the pre-signed download URL once READY. Clients call this on a 15 s poll while the Download Manager is open.

Parameters

Parameter	In	Type	Required	Description	Example
jobId	path	string (UUID)	Yes	Export job identifier returned by <code>POST /api/v1/exports</code> .	f9e8d7c6-...

Response Schema

Field	Type	Description
jobId	string	Job identifier.
status	enum	QUEUED PROCESSING READY FAILED EXPIRED. See the Downloads Manager note above — Figma's UI only shows 3 collapsed badge states; unconfirmed whether that changes this enum.
progressPct	integer [0–100]	Meaningful only when <code>status=PROCESSING</code> .
downloadUrl	string null	Pre-signed GCS URL. Non-null only when READY. Expires 24h post-generation.
expiresAt	ISO 8601 null	Added per UI-Contract-Analysing findings (2026-07). When the pre-signed <code>downloadUrl</code> expires (24h post-generation). Non-null only when <code>status=READY</code> . UI needs this as a structured field to render the "Expired" label without re-deriving it from <code>completedAt</code> .
errorReason	string null	Present when FAILED.
completedAt	ISO 8601 null	Completion timestamp.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/exports/f9e8d7c6-b5a4-3210-fedc-ba9876543210' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "jobId": "f9e8d7c6-b5a4-3210-fedc-ba9876543210",
  "status": "READY",
  "progressPct": 100,
  "downloadUrl": "https://storage.googleapis.com/meridian-exports/f9e8...?X-Goog-Signature=...",
  "expiresAt": "2026-06-23T10:17:43+05:30",
  "errorReason": null,
  "completedAt": "2026-06-22T10:19:22+05:30"
}
```

2. Instock Monitoring — KPI Cards

KPI cards at the top of Dashboard B: 3 headline metrics, a single delta comparator selected via the "Compare metrics vs" radio, worst-FSN modal, and DS < 90% list. All endpoints use infinite scroll — no pagination.

Delta comparator (compareVs): A page-level radio controls which baseline all delta values are computed against. When changed, all endpoints on this page must be re-fetched. Each response returns a single `delta` value — **not** a `deltas` object with 4 keys.

Client-side filtering pattern: All modal endpoints return the complete dataset in one call (no pagination). Any in-modal search, sort, or column filter must be applied client-side — no new API call. A new API call is only required when an uber filter (`fsnCategory` , `fsnPortfolio` , `storeAge` , `storeOpd`) or the `compareVs` radio changes.

"Poor Performing" vs "All" (added per UI-Contract-Analysing findings, 2026-07): Several list views (Worst FSNs here, plus Sub-Pivots and Dark Stores on the City Table page) have a page-level toggle — `Poor Performing (N) / All (M)` — where the UI needs **both** counts up front to render the toggle labels, plus the ability to fetch either scoped set. This is not the same thing as the existing `isCritical` per-row flag (which stays as-is, unrenamed) — it's an additional `filter` request param + `poorPerformingCount` / `allCount` response fields, described inline below wherever it applies.

GET /api/v1/instock/kpi — KPI Cards (3 Headline Metrics)

Returns the three headline KPI values (overall instock %, OOS FSN×DS count, DS below 90% count), each with a single delta value for the selected comparator.

YESTERDAY_START implementation note: Requires J5 to persist the previous day's 06:00 IST snapshot. Returns `null` until that baseline is added.

Request Parameters

Parameter	Type	Required	Description	Example
<code>compareVs</code>	enum	Optional — default: <code>LAST_SNAPSHOT</code>	Comparator baseline for all delta values. Values: <code>LAST_SNAPSHOT</code> <code>DAY_START</code> <code>YESTERDAY_START</code> <code>LAST_WEEK</code> .	<code>DAY_START</code>
<code>fsnCategory</code>	string[] (CSV)	Optional	Uber filter. Values: <code>T100</code> <code>T500</code> <code>T2500</code> <code>T5000</code> <code>T7500</code> <code>Rest</code> <code>Festive</code> <code>Sampling</code> . Default: All.	<code>T100,T500</code>
<code>fsnPortfolio</code>	string[] (CSV)	Optional	Uber filter. Values: <code>Dry</code> <code>DBEFM</code> . Default: All.	<code>Dry</code>
<code>storeAge</code>	string[] (CSV)	Optional	Uber filter. Values: <code>0-15</code> <code>15-30</code> <code>30-60</code> <code>>60</code> (days since launch). Default: All.	<code>>60</code>
<code>storeOpd</code>	string[] (CSV)	Optional	Uber filter. Values: <code><500</code> <code>500-1000</code> <code>1000-1500</code> <code>1500-2000</code> <code>2000-2500</code> <code>2500-3000</code> <code>>3000</code> (orders/day). Default: All.	<code>>3000</code>

Response Schema

Field	Type	Description
asOfTs	ISO 8601	Timestamp of the ATP snapshot used for current values.
compareVs	string	Echoed comparator used for all <code>delta</code> fields.
overallInstockPct.value	number	Current overall instock % across all FSN×DS pairs in scope.
overallInstockPct.delta	number null	Δ vs the selected <code>compareVs</code> baseline. Positive = improvement. Null if baseline unavailable.
oosFsnDsCount.value	integer	Count of FSN×DS pairs currently OOS (ATP=0).
oosFsnDsCount.delta	integer null	Δ vs the selected baseline. Positive = more OOS (regression). Null if baseline unavailable.
dsBelow90Count.value	integer	Dark stores with instock % < configured threshold (default 90%).
dsBelow90Count.threshold	number	Configured threshold (default 90). Included for display.
dsBelow90Count.delta	integer null	Δ vs the selected baseline. Null if baseline unavailable.

Example Request

```
GET /api/v1/instock/kpi?compareVs=DAY_START&fsnCategory=T100,T500
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/kpi?compareVs=DAY_START&fsnCategory=T100,T500' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "asOfTs": "2026-06-22T15:45:00+05:30",
  "compareVs": "DAY_START",
  "overallInstockPct": { "value": 87.4, "delta": 1.2 },
  "oosFsnDsCount": { "value": 14820, "delta": -1540 },
  "dsBelow90Count": { "value": 34, "threshold": 90, "delta": -5 }
}
```

GET /api/v1/instock/kpi/worst-fsns — View Worst FSNs

Returns all FSNs ranked by OOS Breadth % descending. Triggered by the "View Worst FSNs" CTA on the OOS FSN×DS Count KPI card. Infinite scroll — no pagination.

Client-side filtering: Once loaded, filter rows in memory by `bu`, `sc`, `vertical`, `brand`, `topRootCause`, or `productName` text search. No new API call needed — the full FSN list is in the response.

Request Parameters

Parameter	Type	Required	Description	Example
<code>compareVs</code>	enum	Optional — default: <code>LAST_SNAPSHOT</code>	Comparator baseline. Values: <code>LAST_SNAPSHOT</code> <code>DAY_START</code> <code>YESTERDAY_START</code> <code>LAST_WEEK</code> .	<code>DAY_START</code>
<code>filter</code>	enum	Optional — default: <code>ALL</code>	Added per UI-Contract-Analysing findings. Values: <code>POOR_PERFORMING</code> (rows where <code>isCritical=true</code> only) <code>ALL</code> . Response always includes both <code>poorPerformingCount</code> and <code>allCount</code> regardless of which filter is active, so the UI can render both toggle labels without a second call.	<code>POOR_PERFORMING</code>
<code>fsnCategory</code>	string[] (CSV)	Optional	Uber filter.	<code>T100</code>
<code>fsnPortfolio</code>	string[] (CSV)	Optional	Uber filter.	
<code>storeAge</code>	string[] (CSV)	Optional	Uber filter.	
<code>storeOpd</code>	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
<code>poorPerformingCount</code>	integer	Added per UI-Contract-Analysing findings. Count of FSNs with <code>isCritical=true</code> in the current scope, regardless of active <code>filter</code> .
<code>allCount</code>	integer	Added. Total FSN count in the current scope, regardless of active <code>filter</code> .
<code>fsns[].fsn</code>	string	FSN identifier.
<code>fsns[].productName</code>	string	Human-readable product name.
<code>fsns[].oosStoreCount</code>	integer	Dark stores where this FSN is currently OOS.
<code>fsns[].totalStoreCount</code>	integer	Total stores carrying this FSN (denominator).
<code>fsns[].oosBreadthPct</code>	number	$\text{oosStoreCount} / \text{totalStoreCount} \times 100$.
<code>fsns[].instockPct</code>	number	Instock % across all carrying stores.
<code>fsns[].delta</code>	number null	Δ in <code>instockPct</code> vs selected <code>compareVs</code> baseline.
<code>fsns[].topRootCause</code>	string	Dominant root cause: <code>SC_DELAY</code> <code>CRON_UNDERALLOC</code> <code>INACCURATE_NORM</code> <code>STN_DROPOFF</code> .
<code>fsns[].isCritical</code>	boolean	True when <code>oosBreadthPct > threshold</code> (default 50%). This is the field the UI labels "Poor Performing" — no rename, just noting the display-label mapping.

Field	Type	Description
<code>fsns[].otherOosCauses</code>	Array<{label, pct}> null	Added per UI-Contract-Analysing findings (Figma confirmed). Non-IWIT OOS causes, e.g. <code>Direct Inwarding</code> , <code>DS Issues</code> . Null/omitted when none apply (UI renders "--"). Open question: is the label set fixed (enum) or free-form — needs backend confirmation.
<code>fsns[].bu</code>	string	Business Unit — included for client-side filtering, no new API call needed.
<code>fsns[].sc</code>	string	Super Category — included for client-side filtering.
<code>fsns[].vertical</code>	string	Vertical — included for client-side filtering.
<code>fsns[].brand</code>	string	Brand — included for client-side filtering.

Example Request

```
GET /api/v1/instock/kpi/worst-fsns?compareVs=DAY_START&fsnCategory=T100&filter=P00R_PERFORMING
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/kpi/worst-fsns?compareVs=DAY_START&fsnCategory=T100&filter=P00R_PERFORMING' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "poorPerformingCount": 223,
  "allCount": 21623,
  "fsns": [
    {
      "fsn": "ARDET6ZNEMZRRTFK",
      "productName": "Colgate MaxFresh 200g",
      "oosStoreCount": 42, "totalStoreCount": 48,
      "oosBreadthPct": 87.5, "instockPct": 12.5,
      "delta": -2.1, "topRootCause": "SC_DELAY", "isCritical": true,
      "otherOosCauses": [{ "label": "Direct Inwarding", "pct": 2 }],
      "bu": "Personal Care", "sc": "Oral Care",
      "vertical": "Toothpaste", "brand": "Colgate"
    }
  ]
}
```

GET /api/v1/instock/kpi/low-instock-stores — List DS Below 90%

Returns all dark stores where current instock % is below the configured threshold (default 90%), sorted by `instockPct` ascending (worst first). Triggered from the DS below 90% KPI card. Infinite scroll — no pagination.

Client-side filtering: Once loaded, filter rows in memory by `city`, `topRootCause`, or `dsName` text search. No new API call needed.

Request Parameters

Parameter	Type	Required	Description	Example
compareVs	enum	Optional — default: LAST_SNAPSHOT	Comparator baseline. Values: LAST_SNAPSHOT DAY_START YESTERDAY_START LAST_WEEK .	DAY_START
threshold	number	Optional — default: 90	Instock % threshold for filtering.	90
fsnCategory	string[] (CSV)	Optional	Uber filter.	
fsnPortfolio	string[] (CSV)	Optional	Uber filter.	
storeAge	string[] (CSV)	Optional	Uber filter.	
storeOpd	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
stores[].dsCode	string	Dark store / FC code.
stores[].dsName	string	Display name.
stores[].cityCode	string	Split from the old single city field per UI-Contract-Analysing findings — the City dropdown filter in the modal needs a code, not just a display name.
stores[].cityName	string	Split from the old single city field. Display name.
stores[].instockPct	number	Current instock %.
stores[].delta	number null	Δ in instockPct vs selected compareVs baseline.
stores[].oosFsnCount	integer	OOS FSN count at this store (in scope).
stores[].totalFsnCount	integer	Total FSNs in scope at this store.
stores[].topRootCause	string	Dominant root cause bucket.

Example Request

```
GET /api/v1/instock/kpi/low-instock-stores?compareVs=DAY_START&threshold=90
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/kpi/low-instock-stores?compareVs=DAY_START&threshold=90' \
--header 'Authorization: Bearer {sso-token}' \
```

```
--header 'Accept: application/json'
```

Example Response

```
{
  "stores": [
    {
      "dsCode": "BLR_DS_017", "dsName": "HSR Layout DS",
      "cityCode": "BLR", "cityName": "Bengaluru",
      "instockPct": 61.2, "delta": -3.4,
      "oosFsnCount": 187, "totalFsnCount": 483,
      "topRootCause": "CRON_UNDERALLOC"
    }
  ]
}
```

GET /api/v1/instock/fsn/{fsn}/dark-stores — Dark Stores for a Specific FSN

New endpoint, added per UI-Contract-Analysing findings (Figma-confirmed screen: "FSN: [FSN ID] | Dark Stores"). Even with the old Product Drilldown endpoint correctly removed, this specific drill-down — given an FSN, list the dark stores carrying/OOS for it, with live sourcing-hub ATP — was not covered by any existing endpoint. Reached by clicking the "Dark Stores" count on a Worst FSNs modal row.

Request Parameters

Parameter	In	Type	Required	Description	Example
fsn	path	string	Yes	FSN identifier.	FSN0001891081
filter	query	enum	Optional — default: ALL	Values: POOR_PERFORMING ALL . Same toggle pattern as worst-fsns above.	POOR_PERFORMING
fsnCategory , fsnPortfolio , storeAge , storeOpd	query	string[] (CSV)	Optional	Uber filters, same as other Instock Monitoring endpoints.	

Response Schema

Field	Type	Description
poorPerformingCount	integer	Count of dark stores currently OOS for this FSN.
allCount	integer	Total dark stores carrying this FSN.
darkStores[].dsCode	string	Dark store / FC code.
darkStores[].dsName	string	Display name.
darkStores[].instock	boolean	Per Figma, this renders as Yes/No at this granularity, not a %.
darkStores[].sourcingHubAtp	Array<shCode, atpQty>	Live ATP for this FSN at each sourcing hub feeding this DS.
darkStores[].activeIwitCauses	Array<cause, pct>	SC_DELAY CRON_UNDERALLOC INACCURATE_NORM STN_DROP0FF breakdown for this DS.

Field	Type	Description
<code>darkStores[].other0osCauses</code>	<code>Array<{label, pct}> null</code>	Same as worst-fsns — Direct Inwarding / DS Issues, null when none apply.

Example Request

```
GET /api/v1/instock/fsn/FSN0001891081/dark-stores?filter=POOR_PERFORMING
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/fsn/FSN0001891081/dark-stores?filter=POOR_PERFORMING' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "poorPerformingCount": 14,
  "allCount": 160,
  "darkStores": [
    {
      "dsCode": "BOM_DS_026", "dsName": "Bandra DS",
      "inStock": false,
      "sourcingHubAtp": [
        { "shCode": "BOM_SH_BHIWANDI", "atpQty": 56 },
        { "shCode": "BOM_SH_THANE", "atpQty": 45 }
      ],
      "activeIwitCauses": [{ "cause": "SC_DELAY", "pct": 12 }],
      "other0osCauses": null
    }
  ]
}
```


3. Instock Monitoring — City Instock Table & RCA

City instock table, sub-pivot nested rows, dark store sub-table, four RCA modals (B7.1–B7.4), and async export. All table endpoints use infinite scroll — no pagination.

Page-load default — inline dark store sub-rows: On initial page load, `GET /api/v1/instock/cities` is called with `subPivotType=DARK_STORE` (the default). The response embeds dark store rows inline under each city — no separate dark-stores call needed. When the user switches the View Filter to another sub-pivot dimension, `GET /api/v1/instock/cities/{cityCode}/sub-pivots?pivotType=<dimension>` is called per city.

Client-side filtering pattern: All endpoints return the complete dataset in one call (no pagination). In-modal search, sort, and column filters must be applied client-side — no new API call. A new API call is only required when an uber filter or `compareVs` changes.

Product Drilldown endpoint removed: `GET /api/v1/instock/product-drilldown` has been removed in the updated Figma UX. FSN-level drilldown now goes directly to the RCA endpoints with `scope=FSN`.

"Poor Performing" vs "All" (added per UI-Contract-Analysing findings, 2026-07): Sub-Pivots and Dark Stores below now support a `filter` param (`P00R_PERFORMING` | `ALL`) plus `poorPerformingCount` / `allCount` response fields, matching the same toggle pattern on the KPI Cards page's `worst-fsns` and `fsn/{fsn}/dark-stores` endpoints.

Secondary API calls from RCA modals:

SC RCA (B7.1): on STN row click → `GET /api/v1/replenishment/execution/stns/{stnId}/consignments`

SC RCA export → `POST /api/v1/exports` with `exportType=SC_RCA`

Cron RCA export → `POST /api/v1/exports` with `exportType=CRON_RCA`

Norm RCA export → `POST /api/v1/exports` with `exportType=NORM_RCA`

STN Drop-off RCA export → `POST /api/v1/exports` with `exportType=STN_RCA`

GET /api/v1/instock/cities — City Instock Table

Returns one row per city with instock metrics, a single delta value, root cause breakdown, and optionally inline sub-rows for the selected sub-pivot dimension. Default on page load: `subPivotType=DARK_STORE`, embedding dark store rows under each city — no separate API call needed.

Request Parameters

Parameter	Type	Required	Description	Example
<code>compareVs</code>	enum	Optional — default: <code>LAST_SNAPSHOT</code>	Comparator baseline for all delta values. Values: <code>LAST_SNAPSHOT</code> <code>DAY_START</code> <code>YESTERDAY_START</code> <code>LAST_WEEK</code> .	<code>DAY_START</code>
<code>subPivotType</code>	enum	Optional — default: <code>DARK_STORE</code>	Embeds sub-rows under each city. Values: <code>BU</code> <code>SC</code> <code>VERTICAL</code> <code>BRAND</code> <code>SOURCING_HUB</code> <code>NETWORK_COMPONENT</code> <code>DARK_STORE</code> <code>NONE</code> . Pass <code>NONE</code> to suppress sub-rows.	<code>DARK_STORE</code>
<code>fsnCategory</code>	string[] (CSV)	Optional	Uber filter. Values: <code>T100</code> <code>T500</code> <code>T2500</code> <code>T5000</code> <code>T7500</code> <code>Rest</code> <code>Festive</code> <code>Sampling</code> .	<code>T100</code>
<code>fsnPortfolio</code>	string[] (CSV)	Optional	Uber filter. Values: <code>Dry</code> <code>DBEFM</code> .	
<code>storeAge</code>	string[] (CSV)	Optional	Uber filter. Values: <code>0–15</code> <code>15–30</code> <code>30–60</code> <code>>60</code> .	
<code>storeOpd</code>	string[] (CSV)	Optional	Uber filter. Values: <code><500</code> <code>500–1000</code> <code>1000–1500</code> <code>1500–2000</code> <code>2000–2500</code> <code>2500–3000</code> <code>>3000</code> .	

Response Schema — City Row

Field	Type	Description
<code>compareVs</code>	string	Echoed comparator used for all <code>delta</code> fields.
<code>cities[].cityCode</code>	string	City code.
<code>cities[].cityName</code>	string	Display name.

Field	Type	Description
<code>cities[].dsCount</code>	integer	Active dark stores in city.
<code>cities[].totalFsnDsCount</code>	integer	Total active FSN×DS pairs in scope.
<code>cities[].oosFsnDsCount</code>	integer	OOS FSN×DS count.
<code>cities[].instockPct</code>	number	(total – OOS) / total × 100.
<code>cities[].dsBelow90Count</code>	integer	DS where instock % < 90%.
<code>cities[].delta</code>	number null	Δ in <code>instockPct</code> vs selected <code>compareVs</code> baseline.
<code>cities[].topRootCause</code>	string	Dominant bucket label.
<code>cities[].scDelayPct</code>	number	% of OOS attributed to SC Delay. Clickable → B7.1 RCA modal.
<code>cities[].cronUnderallocPct</code>	number	% attributed to Cron Underallocation. Clickable → B7.2.
<code>cities[].inaccurateNormPct</code>	number	% attributed to Inaccurate Norm. Clickable → B7.3.
<code>cities[].stnDropoffPct</code>	number	% attributed to STN Creation Drop-Off. Clickable → B7.4.
<code>cities[].intransitInstockPct</code>	number	% of FSN×DS with effective coverage (in-stock OR open intransit STN). Always ≥ <code>instockPct</code> .
<code>cities[].needsAction</code>	boolean	True when city is top-5 worst OOS or <code>dsBelow90Count</code> > 0.
<code>cities[].otherOosCauses</code>	Array<{label, pct}> null	Added per UI-Contract-Analysing findings (Figma-confirmed "Other OOS Causes" column). Non-IWIT causes, e.g. <code>Direct Inwarding</code> , <code>DS Issues</code> . Null/omitted when none apply (UI renders "--"). Open question: fixed enum or free-form labels — needs backend confirmation.
<code>cities[].subRows[]</code>	Array absent	Present when <code>subPivotType</code> ≠ <code>NONE</code> . Inline sub-pivot rows. See sub-row fields below.

Note: `scDelayPct` + `inaccurateNormPct` + `cronUnderallocPct` + `stnDropoffPct` can exceed `oosFsnDsCount` / `totalFsnDsCount` because one FSN×DS can be attributed to multiple buckets simultaneously.

Response Schema — `subRows[]` (same shape as sub-pivots response rows)

Field	Type	Description
<code>subRows[].pivotType</code>	string	The <code>subPivotType</code> used (echoed per row).
<code>subRows[].entityId</code>	string	Pivot value identifier (BU name / SC name / DS code etc.).
<code>subRows[].entityName</code>	string	Display name.
<code>subRows[].uniqueFsnCount</code>	integer	Distinct FSNs in scope.
<code>subRows[].totalFsnDsCount</code>	integer	Total FSN×DS pairs.
<code>subRows[].oosFsnDsCount</code>	integer	OOS count.
<code>subRows[].instockPct</code>	number	Instock %.
<code>subRows[].dsBelow90Count</code>	integer	DS below 90% (meaningful for SH, Network Component, Dark Store pivots).
<code>subRows[].delta</code>	number null	Δ in <code>instockPct</code> vs selected <code>compareVs</code> baseline.
<code>subRows[].topRootCause</code>	string	Dominant bucket.
<code>subRows[].scDelayPct</code>	number	SC Delay %. Clickable → B7.1.
<code>subRows[].cronUnderallocPct</code>	number	Cron Underallocation %. Clickable → B7.2.
<code>subRows[].inaccurateNormPct</code>	number	Inaccurate Norm %. Clickable → B7.3.
<code>subRows[].stnDropoffPct</code>	number	STN Drop-Off %. Clickable → B7.4.
<code>subRows[].intransitInstockPct</code>	number	Intransit Instock %.
<code>subRows[].otherOosCauses</code>	Array<{label, pct}> null	Added per UI-Contract-Analysing findings. Same shape as <code>cities[].otherOosCauses</code> .

Example Request

```
GET /api/v1/instock/cities?compareVs=DAY_START&subPivotType=DARK_STORE&fsnCategory=T100
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/cities?compareVs=DAY_START&subPivotType=DARK_STORE&fsnCategory=T100' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "compareVs": "DAY_START",
  "cities": [
    {
      "cityCode": "BLR", "cityName": "Bengaluru",
      "dsCount": 52, "totalFsnDsCount": 118640, "oosFsnDsCount": 14920,
      "instockPct": 87.4, "dsBelow90Count": 11, "delta": 1.1,
      "topRootCause": "SC_DELAY",
      "scDelayPct": 42.1, "cronUnderallocPct": 28.5,
      "inaccurateNormPct": 18.3, "stnDropoffPct": 11.1,
      "intransitInstockPct": 93.2, "needsAction": true,
      "otherOosCauses": [{ "label": "Direct Inwarding", "pct": 2 }, { "label": "DS Issues", "pct": 4 }],
      "subRows": [
        {
          "pivotType": "DARK_STORE", "entityId": "BLR_DS_017",
          "entityName": "HSR Layout DS",
          "uniqueFsnCount": 483, "totalFsnDsCount": 483, "oosFsnDsCount": 187,
          "instockPct": 61.3, "dsBelow90Count": 1, "delta": 2.4,
          "topRootCause": "CRON_UNDERALLOC",
          "scDelayPct": 22.6, "cronUnderallocPct": 55.4,
          "inaccurateNormPct": 9.7, "stnDropoffPct": 12.3,
          "intransitInstockPct": 81.0,
          "otherOosCauses": null
        }
      ]
    }
  ]
}
```

GET /api/v1/instock/cities/{cityCode}/sub-pivots — Sub-pivot Rows

Returns the nested breakdown under a city row for the selected View Filter dimension. Called when the user **switches the sub-pivot dimension**. For the default dark store view on page load, use `GET /api/v1/instock/cities` with `subPivotType=DARK_STORE` — no separate call needed.

Request Parameters

Parameter	Type	Required	Description	Example
cityCode	string	Required (path param)	City code.	BLR
pivotType	enum	Required	Sub-pivot dimension. Values: BU SC VERTICAL BRAND SOURCING_HUB NETWORK_COMPONENT DARK_STORE .	SC
filter	enum	Optional — default: ALL	Added per UI-Contract-Analysing findings. Values: POOR_PERFORMING ALL . Response includes both <code>poorPerformingCount</code> and <code>allCount</code> regardless of which is active.	POOR_PERFORMING
compareVs	enum	Optional — default: LAST_SNAPSHOT	Comparator baseline. Values: LAST_SNAPSHOT DAY_START YESTERDAY_START LAST_WEEK .	DAY_START
fsnCategory	string[] (CSV)	Optional	Uber filter.	
fsnPortfolio	string[] (CSV)	Optional	Uber filter.	
storeAge	string[] (CSV)	Optional	Uber filter.	
storeOpd	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
compareVs	string	Echoed comparator.
poorPerformingCount	integer	Added per UI-Contract-Analysing findings.
allCount	integer	Added.
rows[].entityId	string	Pivot value identifier.
rows[].entityName	string	Display name.
rows[].uniqueFsnCount	integer	Distinct FSNs in scope.
rows[].totalFsnDsCount	integer	Total FSN×DS pairs.
rows[].oosFsnDsCount	integer	OOS FSN×DS count.
rows[].instockPct	number	Instock %.
rows[].dsBelow90Count	integer	DS in this entity below 90%.
rows[].delta	number null	Δ in instockPct vs selected compareVs baseline.
rows[].topRootCause	string	Dominant bucket.
rows[].scDelayPct	number	SC Delay %. Clickable → B7.1.
rows[].cronUnderallocPct	number	Cron Underallocation %. Clickable → B7.2.
rows[].inaccurateNormPct	number	Inaccurate Norm %. Clickable → B7.3.
rows[].stnDropoffPct	number	STN Drop-Off %. Clickable → B7.4.
rows[].intransitInstockPct	number	Intransit Instock %.
rows[].otherOosCauses	Array<{label, pct}> null	Added per UI-Contract-Analysing findings.

Example Request

```
GET /api/v1/instock/cities/BLR/sub-pivots?pivotType=SC&compareVs=DAY_START&fsnCategory=T100
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/cities/BLR/sub-pivots?pivotType=SC&compareVs=DAY_START&fsnCategory=T100' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "compareVs": "DAY_START",
  "poorPerformingCount": 4,
  "allCount": 16,
  "rows": [
    {
      "entityId": "Oral Care", "entityName": "Oral Care",
      "uniqueFsnCount": 84, "totalFsnDsCount": 4032, "oosFsnDsCount": 610,
      "instockPct": 84.9, "dsBelow90Count": 8, "delta": 0.9,
      "topRootCause": "CRON_UNDERALLOC",
      "scDelayPct": 31.2, "cronUnderallocPct": 44.8,
      "inaccurateNormPct": 12.4, "stnDropoffPct": 11.6,
      "intransitInstockPct": 91.0,
      "otherOosCauses": [{ "label": "DS Issues", "pct": 3 }]
    }
  ]
}
```

GET /api/v1/instock/cities/{cityCode}/dark-stores — Dark Stores within a Sub-pivot Value

Returns dark store breakdown scoped to a city and optionally a sub-pivot entity. Called when expanding dark stores under a specific sub-pivot row. For the initial page load dark store view, use GET /api/v1/instock/cities with subPivotType=DARK_STORE — no separate call needed.

Request Parameters

Parameter	Type	Required	Description	Example
cityCode	string	Required (path param)	City code.	BLR
compareVs	enum	Optional — default: LAST_SNAPSHOT	Comparator baseline. Values: LAST_SNAPSHOT DAY_START YESTERDAY_START LAST_WEEK .	DAY_START
filter	enum	Optional — default: ALL	Added per UI-Contract-Analysing findings. Values: POOR_PERFORMING ALL . Response includes both poorPerformingCount and allCount regardless of which is active.	POOR_PERFORMING
pivotType	enum	Optional	Set when expanding dark stores under a sub-pivot row. Values: BU SC VERTICAL BRAND SOURCING_HUB NETWORK_COMPONENT .	SC
pivotValue	string	Required when pivotType is set	The specific pivot value to scope dark stores to.	Oral Care
fsnCategory	string[] (CSV)	Optional	Uber filter.	
fsnPortfolio	string[] (CSV)	Optional	Uber filter.	
storeAge	string[] (CSV)	Optional	Uber filter.	
storeOpd	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
compareVs	string	Echoed comparator.
poorPerformingCount	integer	Added per UI-Contract-Analysing findings.
allCount	integer	Added.
darkStores[].dsCode	string	DS FC code.
darkStores[].dsName	string	Display name.
darkStores[].fsnCount	integer	FSNs in scope. (totalFsnDs removed per UI-Contract-Analysing findings — it was always identical to this field for a single DS.)
darkStores[].oosFsnDsCount	integer	OOS count.
darkStores[].instockPct	number	Instock %.
darkStores[].delta	number null	Δ in instockPct vs selected compareVs baseline.
darkStores[].topRootCause	string	Dominant bucket.
darkStores[].scDelayPct	number	SC Delay %.
darkStores[].cronUnderallocPct	number	Cron Underallocation %.
darkStores[].inaccurateNormPct	number	Inaccurate Norm %.
darkStores[].stnDropoffPct	number	STN Drop-Off %.
darkStores[].intransitInstockPct	number	Intransit Instock %.
darkStores[].otherOosCauses	Array<{label, pct}> null	Added per UI-Contract-Analysing findings.

Example Request

```
GET /api/v1/instock/cities/BLR/dark-stores?compareVs=DAY_START&pivotType=SC&pivotValue=Oral%20Care
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/cities/BLR/dark-stores?compareVs=DAY_START&pivotType=SC&pivotValue=Oral%20Care' \
```

```
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "compareVs": "DAY_START",
  "poorPerformingCount": 1,
  "allCount": 1,
  "darkStores": [
    {
      "dsCode": "BLR_DS_017", "dsName": "HSR Layout DS",
      "fsnCount": 84, "oosFsnDsCount": 31,
      "instockPct": 63.1, "delta": 2.4, "topRootCause": "CRON_UNDERALLOC",
      "scDelayPct": 22.6, "cronUnderallocPct": 55.4,
      "inaccurateNormPct": 9.7, "stnDropoffPct": 12.3,
      "intransitInstockPct": 81.0,
      "otherOosCauses": null
    }
  ]
}
```

GET /api/v1/instock/rca/supply-chain — Supply Chain Delay RCA (B7.1)

Returns SC delay breakdown across 10 PRD B7.1 stage labels ranked by impacted quantity. When `expandIssue` is set, all STN rows for that issue are returned inline — no pagination. Export via `POST /api/v1/exports` with `exportType=SC_RCA`.

Client-side filtering on expanded rows: Once `expandIssue` is set and `stns[]` rows are returned, filter them in memory by `sourceFc` (SH), `destinationFc` (DS), `fsn`, or `stnStatus`. No new API call needed. **On STN row click:** Opens consignment detail via `GET /api/v1/replenishment/execution/stns/{stnId}/consignments`.

Request Parameters

Parameter	Type	Required	Description	Example
<code>scope</code>	enum	Required	Trigger context. Values: ALL CITY SUB_PIVOT DARK_STORE FSN.	CITY
<code>cityCode</code>	string	Required when scope=CITY or SUB_PIVOT	City code.	BLR
<code>pivotType</code>	enum	Required when scope=SUB_PIVOT	Sub-pivot dimension.	SC
<code>pivotValue</code>	string	Required when scope=SUB_PIVOT	The specific pivot value.	Oral Care
<code>darkStoreCode</code>	string	Required when scope=DARK_STORE	DS FC code.	BLR_DS_017
<code>fsn</code>	string	Required when scope=FSN	FSN identifier.	ARDET6ZNEMZRRTFK
<code>expandIssue</code>	string	Optional	SC issue label to expand all STN rows for. Must exactly match one of the 10 B7.1 stage labels.	Pending To Actual Dispatch
<code>fsnCategory</code>	string[] (CSV)	Optional	Uber filter.	
<code>fsnPortfolio</code>	string[] (CSV)	Optional	Uber filter.	
<code>storeAge</code>	string[] (CSV)	Optional	Uber filter.	
<code>storeOpd</code>	string[] (CSV)	Optional	Uber filter.	

SC issue labels (attribution priority order): TL not created | Pending To Pick | Picking Lost (INF) | Pending To System Dispatch | Pending To Actual Dispatch | Physical Intransit | Intransit Lost | Pending Destination Acceptance | Pending GRN | QC Rejected at Inwarding.

Response Schema

Field	Type	Description
meta.totalScFsnDsCount	integer	FSNxDS attributed to SC Delay in scope.
meta.scPctOfAllOos	number	SC bucket share of all OOS.
issues[].scIssue	string	One of the 10 B7.1 stage labels.
issues[].fsnDsCount	integer	FSNxDS attributed to this issue.
issues[].impactedQtyUnits	integer	Inventory units attributable to this issue.
issues[].pctOfScOos	number	This issue's share of all SC OOS.
issues[].stnCount	integer	Added per UI-Contract-Analysing findings (Figma-confirmed: "STNs Causing Issue: 23" shown in the collapsed row). Total STN count for this issue, available without needing <code>expandIssue</code> .
issues[].stns	Array null	All STN rows when <code>expandIssue</code> matches this issue. Null otherwise.
issues[].stns[].stnId	string	STN identifier. Clickable → consignment detail.
issues[].stns[].fsn	string	FSN.
issues[].stns[].sourceFc	string	SH FC code. Filter client-side.
issues[].stns[].destinationFc	string	DS FC code. Filter client-side.
issues[].stns[].sku	string	Seller × FSN SKU identifier.
issues[].stns[].qty	integer	Units.
issues[].stns[].stnStatus	string	STN lifecycle status. Filter client-side.
issues[].stns[].stnCreatedAt	ISO 8601	Creation timestamp.
issues[].stns[].dbd	ISO 8601	Dispatch By Date.

Example Request

```
GET /api/v1/instock/rca/supply-chain?scope=CITY&cityCode=BLR&expandIssue=Pending%20To%20Actual%20Dispatch
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/rca/supply-chain?scope=CITY&cityCode=BLR&expandIssue=Pending%20To%20Actual%20Dispatch' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "meta": { "totalScFsnDsCount": 6284, "scPctOfAllOos": 42.1 },
  "issues": [
    {
      "scIssue": "Pending To Actual Dispatch",
      "fsnDsCount": 1840, "impactedQtyUnits": 28400, "pctOfScOos": 29.3,
      "stnCount": 23,
      "stns": [
        {
          "stnId": "STN-BLR-2026-00481234", "fsn": "ARDET6ZNEMZRRTFK",
          "sourceFc": "BLR_SH_01", "destinationFc": "BLR_DS_017",
          "sku": "ARDET6Z_SEL001", "qty": 40, "stnStatus": "CREATED",
          "stnCreatedAt": "2026-06-22T07:30:00+05:30",
          "dbd": "2026-06-22T14:00:00+05:30"
        }
      ]
    },
    { "scIssue": "Physical Intransit", "fsnDsCount": 920, "impactedQtyUnits": 14100, "pctOfScOos": 14.7, "stnCount": 17, "stns": null }
  ]
}
```

GET /api/v1/instock/rca/cron-underallocation — Cron Underallocation RCA (B7.2)

Returns cron underallocation breakdown across 19 constraint labels from the PRD CONSTRAINT_HIERARCHY, ranked by impacted quantity. When `expandIssue` is set, all FSN rows are returned inline — no pagination. Export via `P0ST /api/v1/exports` with `exportType=CRON_RCA`.

Client-side filtering on expanded rows: Once `expandIssue` is set and `fsns[]` rows are returned, filter them in memory by `darkStore`, `fsn`, or `productName` text search. No new API call needed.

Request Parameters

Parameter	Type	Required	Description	Example
scope	enum	Required	Values: ALL CITY SUB_PIVOT DARK_STORE FSN .	CITY
cityCode	string	Required when scope=CITY or SUB_PIVOT	City code.	BLR
pivotType	enum	Required when scope=SUB_PIVOT	Sub-pivot dimension.	
pivotValue	string	Required when scope=SUB_PIVOT	The specific pivot value.	
darkStoreCode	string	Required when scope=DARK_STORE	DS FC code.	
expandIssue	string	Optional	Constraint label to expand all FSN rows for.	INVENTORY_AT_SRC
fsnCategory	string[] (CSV)	Optional	Uber filter.	
fsnPortfolio	string[] (CSV)	Optional	Uber filter.	
storeAge	string[] (CSV)	Optional	Uber filter.	
storeOpd	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
meta.totalCronFsnDsCount	integer	FSN×DS attributed to Cron Underallocation.
meta.cronPctOfAllOos	number	Cron bucket share of all OOS.
issues[].cronIssue	string	One of 19 PRD constraint labels: INVENTORY_AT_SRC , CAPACITY_STORAGE , CASE_SIZE , NDoH Breached , S2S Error , Not Part of Master , etc.
issues[].fsnDsCount	integer	FSN×DS attributed to this constraint.
issues[].designNorm	number	Sum of design norm across all FSN×DS in this bucket.
issues[].impactedQtyUnits	integer	Unallocated units.
issues[].pctOfCronOos	number	Share of cron OOS.
issues[].fsns	Array null	All FSN rows when expandIssue matches. Null otherwise.
issues[].fsns[].fsn	string	FSN.
issues[].fsns[].productName	string	Product name.
issues[].fsns[].darkStore	string	DS code. Filter client-side.
issues[].fsns[].currentIntransitQty	integer	Current intransit quantity across all SHs.
issues[].fsns[].p0ShCronPhase	string	P0 SH cron phase + datetime.
issues[].fsns[].p0ShDesignNorm	number	P0 norm value.
issues[].fsns[].p0ShCronDeficit	number	P0 deficit.
issues[].fsns[].p0ShImpactedQty	integer	Unallocated qty from P0 SH.
issues[].fsns[].p1ShCronPhase	string null	P1 SH cron phase. Null if no P1.

Field	Type	Description
issues[].fsns[].p1ShDesignNorm	number null	P1 norm. Null if no P1.
issues[].fsns[].p1ShCronDeficit	number null	P1 deficit. Null if no P1.
issues[].fsns[].p1ShImpactedQty	integer null	Unallocated qty from P1 SH. Null if no P1.
issues[].fsns[].p1ShCronIssue	string null	P1 constraint label. Null if no P1.

Example Request

```
GET /api/v1/instock/rca/cron-underallocation?scope=CITY&cityCode=BLR&expandIssue=INVENTORY_AT_SRC
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/rca/cron-underallocation?scope=CITY&cityCode=BLR&expandIssue=INVENTORY_AT_SRC' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "meta": { "totalCronFsnDsCount": 4270, "cronPctOfAll0os": 28.5 },
  "issues": [
    {
      "cronIssue": "INVENTORY_AT_SRC",
      "fsnDsCount": 1420, "designNorm": 98400, "impactedQtyUnits": 31200, "pctOfCron0os": 33.3,
      "fsns": [
        {
          "fsn": "LNKXYZ123", "productName": "Lays Classic Salted 26g",
          "darkStore": "BLR_DS_017", "currentIntransitQty": 0,
          "p0ShCronPhase": "Phase 2 | 2026-06-22 08:00",
          "p0ShDesignNorm": 150, "p0ShCronDeficit": 120, "p0ShImpactedQty": 120,
          "p1ShCronPhase": null, "p1ShDesignNorm": null, "p1ShCronDeficit": null,
          "p1ShImpactedQty": null, "p1ShCronIssue": null
        }
      ]
    }
  ]
}
```

GET /api/v1/instock/rca/inaccurate-norm — Inaccurate Norm RCA + MAPE Export (B7.3)

Returns forecast error breakdown across 5 MAPE bands. When `expandBucket` is set, all FSN rows are returned inline — no pagination. Export via `POST /api/v1/exports` with `exportType=NORM_RCA`.

Client-side filtering on expanded rows: Once `expandBucket` is set and `fsns[]` rows are returned, filter them in memory by `darkStore`, `fsn`, or `productName` text search. No new API call needed.

instockCorrectedDrr is null in V1 — requires 14-day hourly ATP history. Frontend renders `--` with a tooltip until ATP history is backfilled.

Request Parameters

Parameter	Type	Required	Description	Example
scope	enum	Required	Values: ALL CITY SUB_PIVOT DARK_STORE FSN .	CITY
cityCode	string	Required when scope=CITY or SUB_PIVOT	City code.	BLR
pivotType	enum	Required when scope=SUB_PIVOT	Sub-pivot dimension.	
pivotValue	string	Required when scope=SUB_PIVOT	The specific pivot value.	

Parameter	Type	Required	Description	Example
darkStoreCode	string	Required when scope=DARK_STORE	DS FC code.	
expandBucket	string	Optional	MAPE band to expand. Values: 100–125 125–150 150–175 175–200 >200 .	125–150
fsnCategory	string[] (CSV)	Optional	Uber filter.	
fsnPortfolio	string[] (CSV)	Optional	Uber filter.	
storeAge	string[] (CSV)	Optional	Uber filter.	
storeOpd	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
meta.totalNormFsnDsCount	integer	FSN×DS attributed to Inaccurate Norm.
meta.normPctOfAllOos	number	Norm bucket share of all OOS.
mapeBuckets[].mapeBucket	string	Band: 100–125% 125–150% 150–175% 175–200% >200% .
mapeBuckets[].fsnDsCount	integer	FSN×DS where (actual_demand / P0_norm × 100) falls in this band.
mapeBuckets[].pctOfDemandOos	number	Band's share of all norm OOS.
mapeBuckets[].fsns	Array null	All FSN rows for expandBucket. Null when not expanded.
mapeBuckets[].fsns[].fsn	string	FSN.
mapeBuckets[].fsns[].productName	string	Product name.
mapeBuckets[].fsns[].darkStore	string	DS code. Filter client-side.
mapeBuckets[].fsns[].p0CronTime	ISO 8601	P0 cron runtime.
mapeBuckets[].fsns[].p0Norm	number	P0 replenishment norm (units/day).
mapeBuckets[].fsns[].p0SafetyStock	number	P0 SH safety stock at cron time.
mapeBuckets[].fsns[].p0CycleStock	number	P0 SH cycle stock at cron time.
mapeBuckets[].fsns[].p0ForecastedLt	number	P0 SH design lead time (hours).
mapeBuckets[].fsns[].instockCorrectedDrr	number null	Historical DRR corrected for OOS periods. Null in V1.
mapeBuckets[].fsns[].forecastedDrr	number	Norm-implied daily demand rate.
mapeBuckets[].fsns[].actualSalesFromP0CronTime	number	Realised sales from P0 cron time to now.
mapeBuckets[].fsns[].mapePct	number	(actualSales – p0Norm) / p0Norm × 100.

Example Request

```
GET /api/v1/instock/rca/inaccurate-norm?scope=CITY&cityCode=BLR&expandBucket=125-150
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/rca/inaccurate-norm?scope=CITY&cityCode=BLR&expandBucket=125-150' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "meta": { "totalNormFsnDsCount": 2740, "normPctOfAllOos": 18.3 },
  "mapeBuckets": [
    {
      "mapeBucket": "125-150%", "fsnDsCount": 1180, "pctOfDemandOos": 43.1,
```

```
    "fsns": [
      {
        "fsn": "MFXABC789", "productName": "Maggi Noodles Masala 70g",
        "darkStore": "BLR_DS_022",
        "p0CronTime": "2026-06-22T06:00:00+05:30",
        "p0Norm": 8.0, "p0SafetyStock": 2.0, "p0CycleStock": 6.0, "p0ForecastedLt": 8,
        "instockCorrectedDrr": null, "forecastedDrr": 8.0,
        "actualSalesFromP0CronTime": 12.0, "mapePct": 50.0
      }
    ]
  }
}
```

GET /api/v1/instock/rca/stn-creation-dropoff — STN Creation Drop-Off RCA (B7.4)

Returns STN creation failure breakdown across 35 PRD failure stages, ranked by impacted quantity. When `expandStage` is set, all TR/STR rows for that stage are returned inline — no pagination. Export via `POST /api/v1/exports` with `exportType=STN_RCA`.

Client-side filtering on expanded rows: Once `expandStage` is set and `trs[]` rows are returned, filter them in memory by `sourceFc` (SH), `destinationFc` (DS), `fsn`, or `trStatus`. No new API call needed.

Request Parameters

Parameter	Type	Required	Description	Example
<code>scope</code>	enum	Required	Values: <code>ALL</code> <code>CITY</code> <code>SUB_PIVOT</code> <code>DARK_STORE</code> <code>FSN</code> .	<code>CITY</code>
<code>cityCode</code>	string	Required when <code>scope=CITY</code> or <code>SUB_PIVOT</code>	City code.	<code>BLR</code>
<code>pivotType</code>	enum	Required when <code>scope=SUB_PIVOT</code>	Sub-pivot dimension.	
<code>pivotValue</code>	string	Required when <code>scope=SUB_PIVOT</code>	The specific pivot value.	
<code>darkStoreCode</code>	string	Required when <code>scope=DARK_STORE</code>	DS FC code.	
<code>expandStage</code>	integer	Optional	Stage number (1–35) to expand all TR-level rows for.	<code>26</code>
<code>fsnCategory</code>	string[] (CSV)	Optional	Uber filter.	
<code>fsnPortfolio</code>	string[] (CSV)	Optional	Uber filter.	
<code>storeAge</code>	string[] (CSV)	Optional	Uber filter.	
<code>storeOpd</code>	string[] (CSV)	Optional	Uber filter.	

Response Schema

Field	Type	Description
<code>meta.totalStnFsnDsCount</code>	integer	FSN×DS attributed to STN Creation Drop-Off.
<code>meta.stnPctOfAllOos</code>	number	STN bucket share of all OOS.
<code>stages[].stageNum</code>	integer	Stage number from PRD B7.4 catalog (1–35).
<code>stages[].stageLabel</code>	string	Full label, e.g. "Requirement is not in multiple of case size".
<code>stages[].fsnDsCount</code>	integer	FSN×DS attributed to this stage.
<code>stages[].impactedQtyUnits</code>	integer	Allocated units that could not become STNs at this stage.
<code>stages[].pctOfStnOos</code>	number	Stage's share of STN OOS.
<code>stages[].trs</code>	Array null	All TR rows when <code>expandStage</code> matches this stage. Null otherwise.
<code>stages[].trs[].trStrId</code>	string	Transfer Request / STR identifier.
<code>stages[].trs[].fsn</code>	string	FSN.
<code>stages[].trs[].sourceFc</code>	string	SH FC code. Filter client-side.
<code>stages[].trs[].destinationFc</code>	string	DS FC code. Filter client-side.
<code>stages[].trs[].qty</code>	integer	Allocated units.

Field	Type	Description
stages[].trs[].trStatus	string	TR lifecycle status. Filter client-side.
stages[].trs[].trCreatedAt	ISO 8601	Creation timestamp.
stages[].trs[].dbd	ISO 8601	Dispatch By Date.

Example Request

```
GET /api/v1/instock/rca/stn-creation-dropoff?scope=CITY&cityCode=BLR&expandStage=26
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/rca/stn-creation-dropoff?scope=CITY&cityCode=BLR&expandStage=26' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "meta": { "totalStnFsnDsCount": 1660, "stnPctOfAllOos": 11.1 },
  "stages": [
    {
      "stageNum": 26, "stageLabel": "Requirement is not in multiple of case size",
      "fsnDsCount": 480, "impactedQtyUnits": 7200, "pctOfStnOos": 28.9,
      "trs": [
        {
          "trStrId": "TR-2026-BLR-00192841", "fsn": "ARDET6ZNEMZRRTFK",
          "sourceFc": "BLR_SH_01", "destinationFc": "BLR_DS_017",
          "qty": 25, "trStatus": "REJECTED",
          "trCreatedAt": "2026-06-22T08:15:00+05:30",
          "dbd": "2026-06-22T14:00:00+05:30"
        }
      ]
    }
  ]
}
```

POST /api/v1/instock/export — Export All OOS Report (Async)

Triggers an async full OOS export at the selected scope. Output: 31-column CSV (one row per OOS FSNxDS pair, B6 schema). Poll via `GET /api/v1/exports/{jobId}`.

Request Parameters

Parameter	Type	Required	Description	Example
scope	enum	Required	Export scope. Values: ALL CITY SUB_PIVOT DARK_STORE .	CITY
cityCode	string	Required when scope=CITY or SUB_PIVOT	City code.	BLR
pivotType	enum	Required when scope=SUB_PIVOT	Sub-pivot dimension.	SC
pivotValue	string	Required when scope=SUB_PIVOT	Pivot value.	Oral Care
darkStoreCode	string	Required when scope=DARK_STORE	DS FC code.	
fsnCategory	string[] (CSV)	Optional	Uber filter. Values: T100 T500 T2500 T5000 T7500 Rest Festive Sampling .	T100,T500
fsnPortfolio	string[] (CSV)	Optional	Uber filter. Values: Dry DBEFM .	
storeAge	string[] (CSV)	Optional	Uber filter. Values: 0-15 15-30 30-60 >60 .	

Parameter	Type	Required	Description	Example
storeOpd	string[] (CSV)	Optional	Uber filter. Values: <500 500–1000 1000–1500 1500–2000 2000–2500 2500–3000 >3000 .	

Response Schema

Field	Type	Description
jobId	string (UUID)	Poll via GET /api/v1/exports/{jobId} .
status	string	Always QUEUED at submission.
estimatedRows	integer	Estimated OOS FSN×DS count.
estimatedSeconds	integer	Estimated generation time in seconds.

Example Request

```
POST /api/v1/instock/export
{ "scope": "CITY", "cityCode": "BLR", "fsnCategory": ["T100", "T500"]} }
```

Curl

```
curl --location 'http://{instockctl-host}/api/v1/instock/export' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Content-Type: application/json' \
--data-raw '{"scope":"CITY","cityCode":"BLR","fsnCategory":["T100","T500"],"fsnPortfolio":[],"storeAge":[],"storeOpd":[]}'
```

Example Response

```
{
  "jobId": "c3b2a1d0-e9f8-7654-3210-abcdef012345",
  "status": "QUEUED",
  "estimatedRows": 22800,
  "estimatedSeconds": 60
}
```

4. Instock Diagnostics

FSN × DS and FSN × SH standalone diagnostics, consolidated report export (single and bulk), and dynamic pivot view for Dashboard C.

Open questions from the Figma reads (2026-07) — flagged, not guessed:

1. **"Dynamic Pivot View" tab** was not captured in either Figma pass. The `pivot` endpoint section further below on this page predates both reads and has **not** been reconciled — treat it as unconfirmed until a dedicated Figma read is done. (FSN×SH, previously also unconfirmed here, is now fully specified in Section 1a below via a second Figma read.)
2. **FSN dropdown** appeared to show repeating entries in the source screenshot — unconfirmed whether that's real mock-data duplication or a render/pagination artifact.
3. **"View Details" links** on the 4 root-cause summary cards are assumed to scroll to the matching report section below (all 4 reports render inline on the same page, not as modals) — not confirmed as a scroll-vs-modal behaviour.
4. **Root cause set (4 causes)** — unclear if this is a fixed set of exactly 4, always shown regardless of status, or a dynamic list that could vary in count/type per FSN×DS. Modelled as an array to accommodate either.
5. **STN status enum inconsistency** — this page's drill-down modals show `CLOSED` | `SCHEDULED` | `CANCELLED` | `DISPATCHED` ; Instock Monitoring / Replenishment Execution show `CREATED` | `AT_DS` | `INWARDED` | `DISPATCHED` | `IN_TRANSIT` . These likely need one unified enum — flagging for backend/data team confirmation, not merging silently.
6. **TR IDs repeating** across rows in the source screenshots (same ID on every row) looked like a mock-data placeholder, not a real pattern — don't build against it.
7. **Design Lead Time** shown as a static header value ("4 hours") — unclear if configurable per FSN / Dark Store / Sourcing Hub, or a global constant.
8. **Two overlapping bulk-export specs now exist** — see the panel under "Consolidated Report Download" below. Needs a product/backend decision on whether they're the same feature described twice or genuinely two different triggers.
9. **Downloads Manager entry point** — not visible in the left icon rail in any screenshot so far; may be a top-bar icon or a post-submit link. Not confirmed. The Downloads Manager table itself is documented on **page 1 (Generic)**, reconciled against the existing export job list there.
10. **Query Scope list completeness** — only 2 scopes shown for Consolidated Report Download (`FSN(s)→Dark Store(s)` , `FSN(s)→Sourcing Hub(s)`); there may be more given how many pivot dimensions exist elsewhere — not confirmed.
11. **"Include complete RCA metadata" toggle** — visible in the UI, its effect on actual export columns is not confirmed.
12. **Upload template file format** and **sync-vs-async threshold** for small exports — not confirmed.

GET /api/v1/diagnostics/fsn-ds — FSN × DS Diagnostic

Full OOS diagnostic for a specific FSN × Dark Store pair: header metadata, 4 root-cause status cards, and 4 inline detail reports (Inaccurate Norm, Cron Under-Allocation, STN Creation Drop-off, Supply Chain Execution). Only valid for OOS pairs (ATP = 0); returns `isOos=false` with only header fields populated otherwise.

Rewritten 2026-07 per Figma "Product View (FSN x DS)" build prompt. Supersedes the earlier `rootCauseTags` / `cronAllocationReport` / `stnTimeline` shape — see version history if you need the prior contract. The STN Creation Drop-off Report and Supply Chain Execution Report are new; they didn't exist in any prior version of this contract.

Parameters

Parameter	In	Type	Required	Description	Example
<code>fsn</code>	query	string	Yes	FSN identifier.	<code>FSN0006716</code>

Parameter	In	Type	Required	Description	Example
darkStoreCode	query	string	Yes	Dark Store / FC code.	BLR_DS_001

Response Schema

Field	Type	Description
fsn.fsnId / fsn.title / fsn.brand / fsn.vertical	string	Header metadata — new, not in the prior contract.
darkStore.dsCode	string	New.
darkStore.cityCode / darkStore.cityName	string	New. Figma only showed the display name ("Bengaluru") — cityCode added for consistency with the rest of this contract; confirm instockctl actually has it at this join level.
oosStatus.isOos	boolean	True when ATP=0. If false, only header fields are populated.
oosStatus.atp	integer	New — shown in the header badge, e.g. "OOS (ATP:0)".
lastUpdated	ISO 8601	New.
designLeadTimeHours	number	New. See open question 7 above re: configurability.
causes[].key	string	INACCURATE_NORM CRON_UNDERALLOC STN_DROP0FF SC_DELAY . Replaces the old flat rootCauseTags list — see open question 4 re: whether this set is fixed.
causes[].status	string	ACTIVE INACTIVE . Unlike the old rootCauseTags (active-only list), all causes are returned regardless of status.
causes[].description	string	New. Server-computed one-line diagnostic text.
inaccurateNormReport.durationFrom / durationTo	ISO 8601	Window start = last In-Stock Cron time; end = now.
inaccurateNormReport.elapsedHours	number	New.
inaccurateNormReport.drrPerHour	number	New. Anticipated sales = drrPerHour × elapsedHours .
inaccurateNormReport.anticipatedSales / actualSales	number	Same as prior contract.
inaccurateNormReport.mapePct	number	anticipated − actual / anticipated × 100 .
inaccurateNormReport.thresholdPct	number	New — threshold MAPE is compared against to decide if this cause is ACTIVE .
cronUnderAllocationReport[].sourcingHubCode	string	One row per Sourcing Hub — latest/current cron only. Full 14-run history is a separate drill-down, see below.

Field	Type	Description
<code>cronUnderAllocationReport[].cronDatetime</code>	ISO 8601	
<code>cronUnderAllocationReport[].target / atp / inTransitQty</code>	integer	Same as prior <code>inventoryTarget / atp / cronIntransitAllSh</code> , renamed for consistency with the drill-down endpoint below.
<code>cronUnderAllocationReport[].deficitQty</code>	integer	<code>target - atp - inTransitQty</code> , floored at 0.
<code>cronUnderAllocationReport[].allocatedIwitQty</code>	integer	Same as prior <code>allocatedIwitQtyAllSh</code> .
<code>cronUnderAllocationReport[].unallocatedQty</code>	integer	New — <code>deficitQty - allocatedIwitQty</code> .
<code>cronUnderAllocationReport[].cronConstraints[].type / .qty</code>	string / integer	Replaces the prior single string field <code>cronConstraintsFromSh</code> .
<code>stnCreationDropoffReport.rows[] (per SH) / .total</code>	object	New report. <code>expectedTrs/expectedUnits</code> , <code>actualTrs/actualUnits</code> , <code>dropoffTrs/dropoffUnits</code> , <code>pendingTrs/pendingUnits</code> .
<code>supplyChainExecutionReport.rows[] (per SH) / .total</code>	object	New report. <code>expectedUnits</code> , <code>landedUnits</code> , <code>delayUnits</code> , <code>pendingUnits</code> .
<code>warnings</code>	string[] null	Data quality warnings. Unchanged.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds?fsn=FSN0006716&darkStoreCode=BLR_DS_001' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "fsn": { "fsnId": "FSN0006716", "title": "Fortune Edible Oil", "brand": "Fortune", "vertical": "Edible Oil" },
  "darkStore": { "dsCode": "BLR_DS_001", "cityCode": "BLR", "cityName": "Bengaluru" },
  "oosStatus": { "isOos": true, "atp": 0 },
  "lastUpdated": "2026-06-24T20:03:00+05:30",
  "designLeadTimeHours": 4,
  "causes": [
    { "key": "INACCURATE_NORM", "status": "ACTIVE", "description": "..." },
    { "key": "CRON_UNDERALLOC", "status": "ACTIVE", "description": "..." },
    { "key": "STN_DROPPOFF", "status": "ACTIVE", "description": "..." },
    { "key": "SC_DELAY", "status": "INACTIVE", "description": "..." }
  ],
  "inaccurateNormReport": { "durationFrom": "2026-05-07T06:03:00+05:30", "durationTo": "2026-05-07T11:30:00+05:30", "elapsedHours": 5.45, "drrPerHour": 3, "anticipatedSales": 16, "actualSales": 24, "mapePct": 33, "thresholdPct": 25 },
  "cronUnderAllocationReport": [
    { "sourcingHubCode": "BLR_SH_WHITEFIELD", "cronDatetime": "2026-06-24T20:03:00+05:30", "target": 580, "atp": 0, "inTransitQty": 14, "deficitQty": 566, "allocatedIwitQty": 0, "unallocatedQty": 566, "cronConstraints": [{ "type": "SAFETY_STOCK", "qty": 45 }, { "type": "EXCESS", "qty": 11 }, { "type": "CAP_CONSTRAINT", "qty": 521 } ] }
  ],
  "stnCreationDropoffReport": { "rows": [{ "sourcingHubCode": "BLR_SH_WHITEFIELD", "expectedTrs": 18, "expectedUnits": 324, "actualTrs": 14, "actualUnits": 252, "dropoffTrs": 4, "dropoffUnits": 82, "pendingTrs": 3, "pendingUnits": 24 }], "total": { "expectedTrs": 32, "expectedUnits": 556, "actualTrs": 28, "actualUnits": 512, "dropoffTrs": 4, "dropoffUnits": 82, "pendingTrs": 2, "pendingUnits": 5 } },
  "supplyChainExecutionReport": { "rows": [{ "sourcingHubCode": "BLR_SH_WHITEFIELD", "expectedUnits": 228, "landedUnits": 0, "delayUnits": 29, "pendingUnits": 199 }], "total": { "expectedUnits": 540, "landedUnits": 282, "delayUnits": 31, "pendingUnits": 227 } },
}
```



```
"warnings": ["DRR_HISTORY_UNAVAILABLE"]
}
```

GET /api/v1/diagnostics/fsn-ds/cron-runs — Last 14 Cron Runs (per SH)

Backs the "[Sourcing Hub] I Last 14 Cron Runs" modal. Leaf — no further drill-down.

Params: `fsn` , `darkStoreCode` , `sourcingHubCode` — all required.

Response: `runs[]` with `cronDatetime`, `phase` (P1|P2), `phaseStatus` (IN_WINDOW|ELAPSED), `elapsedStatus` (display string), the same `target/atp/inTransitQty/deficitQty/allocatedIwitQty/unallocatedQty/cronConstraints[]` shape as the inline report, plus `isLatestCron` , `isInstockCron` (booleans; unconfirmed whether both can be true on different rows simultaneously).

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/cron-runs?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/diagnostics/fsn-ds/stn-creation-dropoff-reasons — Drop-off Reasons (per SH)

Backs the "[Sourcing Hub] I STN Creation Drop-off Reasons" modal.

Params: `fsn` , `darkStoreCode` , `sourcingHubCode` — all required.

Response: `reasons[].reason/trCount/qty` + `total.trCount/qty` . "View TRs" → `trs` endpoint below with `reason` set.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/stn-creation-dropoff-reasons?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/diagnostics/fsn-ds/stn-creation-timeline — Last 14 Days STN Creation Timeline (per SH)

Params: `fsn` , `darkStoreCode` , `sourcingHubCode` .

Response: `days[].date/status`
(MAJORITY_DROPPOFF|PARTIAL_DROPPOFF|NO_DROPPOFF|NO_TR_PUSHED)/expectedTrs/expectedUnits/convertedTrs/convertedUnits/droppedTrs/droppedUnits .
"View TRs" → `trs` endpoint below with `date` set.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/stn-creation-timeline?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/diagnostics/fsn-ds/trs — TR List (per reason OR per date)

Shared by "View TRs" from a reason row (pass `reason`) or a timeline date row (pass `date` , response additionally includes `failureReason`). Exactly one of `reason` / `date` expected.

Params: `fsn` , `darkStoreCode` , `sourcingHubCode` , one of `reason` / `date` .

Response: `trs[].trId/status` (CREATED_TO_STN|PENDING|FAILED)/createdAt/dbd/createdBy/sku/qty/failureReason? .

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/trs?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD&reason=Product%20Attr%20Missing' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/diagnostics/fsn-ds/intransit-dropoff-reasons — In-transit Drop-off Reasons (per SH)

Backs the "[Sourcing Hub] | In transit Drop-off Reasons" modal. Same params as `stn-creation-dropoff-reasons`. Response: `reasons[]`. `reason/stnCount/qty`. "View STNs" → `stns` endpoint below with `reason` set.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/intransit-dropoff-reasons?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/diagnostics/fsn-ds/stn-execution-timeline — Last 14 Days STN Execution Timeline (per SH)

Same params as `stn-creation-timeline`. Response: `days[]`. `date/status` (`PENDENCY_GT_DESIGN_LT|INWARDING_PENDING|FULLY_INWARDED|NO_STN_CREATED`)/`expectedQty/inwardedQty`. "View STNs" → `stns` endpoint below with `date` set.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/stn-execution-timeline?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/diagnostics/fsn-ds/stns — STN List (per reason OR per date)

Shared by "View STNs" from an in-transit reason row or an execution timeline date row. Row click → consignment detail, reusing `GET /api/v1/replenishment/execution/stns/{stnId}/consignments` (page 5) — no new consignment endpoint.

Params: `fsn`, `darkStoreCode`, `sourcingHubCode`, one of `reason` / `date`.

Response: `stns[]`. `stnId/status/createdAt/dbd/ageHrs/sku/sellerId/qty`. `status` values seen: `CLOSED|SCHEDULED|CANCELLED|DISPATCHED` — see open question 5.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-ds/stns?
fsn=FSN0006716&darkStoreCode=BLR_DS_001&sourcingHubCode=BLR_SH_WHITEFIELD&date=2026-06-24' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Consignment detail note: the leaf modal here shows a `Qty / Stage` column on the consignment detail endpoint. This is now a **confirmed** field (`consignments[].stage`) on page 5 — added there after two independent Figma reads showed it.

Section 1a — GET /api/v1/diagnostics/fsn-sh — FSN x SH Diagnostic

Rewritten 2026-07 — confirmed via a second Figma read ("Product View (FSN x SH)" tab). This is an aggregate rollup across every Dark Store a Sourcing Hub serves for one FSN, with a per-DS breakdown that links out to the FSNxDS single-store report above (Section 1) for detail —

not a standalone diagnostic report of its own. Supersedes the previous unconfirmed stub for this endpoint.

Parameters

Parameter	In	Type	Required	Description	Example
fsn	query	string	Yes	FSN identifier.	FSN0006716
sourcingHubCode	query	string	Yes	Sourcing Hub FC code.	BLR_SH_KOLAR

Response Schema

Field	Type	Description
fsn.fsnId / .title / .brand / .vertical	string	Same header shape as the FSNxDS endpoint.
sourcingHub.sourcingHubCode / .cityName / .dsCoverage	string / string / integer	dsCoverage = total Dark Stores this SH serves for this FSN.
lastUpdated	ISO 8601	
inStockPct	number	KPI card 1.
oosFsnDs.count / .total	integer	KPI card 2, shown as "2/9".
shAtp	integer	KPI card 3 — this SH's own ATP for this FSN.
oosRootCauseCounts[].cause / .count	string / integer	KPI card 4 chips — count of Dark Stores (under this SH) where each cause is active. Same 4 cause keys as Section 1.
darkStores[].dsCode / .cityName	string	
darkStores[].inStockStatus	enum	IN_STOCK OOS
darkStores[].activeCauses	string[]	Subset of the 4 cause keys, active for this specific Dark Store.

Cross-navigation, not a new endpoint: "View Details" on a Dark Store row navigates client-side to Section 1's fsn-ds report, pre-filled with this fsn + that row's dsCode . No separate contract needed for that transition.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-sh?fsn=FSN0006716&sourcingHubCode=BLR_SH_KOLAR' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "fsn": { "fsnId": "FSN0006716", "title": "Fortune Edible Oil", "brand": "Fortune", "vertical": "Edible Oil" },
  "sourcingHub": { "sourcingHubCode": "BLR_SH_KOLAR", "cityName": "Bengaluru", "dsCoverage": 9 },
```

```

"lastUpdated": "2026-06-24T20:03:00+05:30",
"inStockPct": 79.6,
"oosFsnDs": { "count": 2, "total": 9 },
"shAtp": 13,
"oosRootCauseCounts": [
  { "cause": "SC_DELAY", "count": 1 }, { "cause": "CRON_UNDERALLOC", "count": 1 },
  { "cause": "STN_DROPPOFF", "count": 1 }, { "cause": "INACCURATE_NORM", "count": 1 }
],
"darkStores": [
  { "dsCode": "BLR_DS_027", "cityName": "Hyderabad", "inStockStatus": "OOS", "activeCauses": ["SC_DELAY", "CRON_UNDERALLOC", "STN_DROPPOFF"] },
  { "dsCode": "BLR_DS_028", "cityName": "Bengaluru", "inStockStatus": "IN_STOCK", "activeCauses": ["INACCURATE_NORM", "CRON_UNDERALLOC"] }
]
}

```

GET /api/v1/diagnostics/fsn-options / dark-store-options / sh-options — Search Dropdowns

Backs the searchable FSN / Dark Store / Sourcing Hub selectors on both Product View tabs and the Consolidated Report Download modal below.

Likely partial duplication with the existing shared GET /api/v1/filters/options (page 1): Dark Stores and Sourcing Hubs are already fully enumerable there (sections=darkStores / sections=sourcingHubs), so those two selectors probably don't need new dedicated endpoints — a client-side search box over the existing full list would work, matching the "client-side filtering" pattern used everywhere else in this contract. **FSNs are not** — the catalog is too large to return in full, so fsn-options below is likely genuinely new and needed. Recommend confirming with backend before building 3 new endpoints when 2 may be redundant.

GET /api/v1/diagnostics/fsn-options

Parameter	In	Type	Required	Description
search	query	string	No	Free-text match on FSN id or title.

Response: { items: [{ fsnId, title }] }. Open question 2 above flags a possible duplicate-entries artifact seen in the source screenshot — confirm against live data, not the mock.

Curl

```

curl --location 'http://{instockctl-host}/api/v1/diagnostics/fsn-options?search=fortune' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'

```

Consolidated Report Download — Modal + Downloads Manager

This likely overlaps with the existing "Consolidated Bulk Export" endpoint further below on this page — flagging, not merging. Both let a user pick multiple FSNs plus multiple DS/SH and trigger one bulk export. Differences observed: this one (a) is triggered from the "↓ Consolidated Report" dropdown on both Product View tabs, (b) only supports 2 explicit two-list scopes (FSN(s)→DS(s) , FSN(s)→SH(s)) rather than the older single-list scopes A/B/C, (c) supports an Upload File input mode as an alternative to searching, and (d) has an "Include complete RCA metadata" toggle not present on the older endpoint. This needs a product/backend decision on whether it's the same feature re-specified, a replacement, or a genuinely separate trigger — do not assume either way.

POST /api/v1/diagnostics/consolidated-report — Trigger Export

Field	Type	Required	Description
queryScope	enum	Yes	FSN_TO_DARK_STORE FSN_TO_SOURCING_HUB .
fsnIds	string[]	One of fsnIds / fsnUploadFileId required	From search mode.
fsnUploadFileId	string	See above	From Upload File mode — see upload flow below.
darkStoreIds / darkStoreUploadFileId	string[] / string	Required when queryScope=FSN_TO_DARK_STORE — one of the two.	
sourcingHubIds / sourcingHubUploadFileId	string[] / string	Required when queryScope=FSN_TO_SOURCING_HUB — one of the two.	
includeCompleteRcaMetadata	boolean	No — default true	Effect on output columns not confirmed — see open question 11.

Response: { requestId: "EXP-KD54MZAY", status: "QUEUED" }. Immediately creates a row in the Downloads Manager (page 1). See page 1 for the requestId format discrepancy this raises against the existing export job list.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/consolidated-report' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Content-Type: application/json' \
--data-raw '{
  "queryScope": "FSN_TO_SOURCING_HUB",
  "fsnIds": ["FSN0010011", "FSN0010012"],
  "sourcingHubIds": ["BLR_SH_001", "BLR_SH_002"],
  "includeCompleteRcaMetadata": true
}'
```

Upload File mode — new capability, not covered anywhere else in this contract

Each of the FSN / Dark Store / Sourcing Hub selectors can switch from Search to Upload File mode: user uploads a CSV of ids instead of picking from a dropdown.

Endpoint	Description
GET /api/v1/diagnostics/upload-template? type=FSN DARK_STORE SOURCING_HUB	Returns the CSV template file. Actual column structure not yet seen — see open question 11.
POST /api/v1/diagnostics/upload? type=FSN DARK_STORE SOURCING_HUB (multipart)	Validates and stores the uploaded file. Response: { uploadFileId, fileName, validCount, status: "SUCCESS" "FAILED", errors?: [{ row, message }], errorFileDownloadUrl? }. On success, uploadFileId is passed as fsnUploadFileId /etc. in the trigger call above. On failure, UI shows the row-level errors and offers the error file for download; "Retry" re-opens the file picker — not confirmed whether retry expects a fresh file or reuses the same one.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/upload?type=FSN' \
--header 'Authorization: Bearer {sso-token}' \
--form 'file=@fsns.csv'
```

Downloads Manager table itself lives on page 1 (Generic — Filters, Status & Exports), reconciled there against the existing `GET /api/v1/exports` job list. This section only covers the trigger + upload flow that's specific to Instock Diagnostics.

POST /api/v1/diagnostics/consolidated/export — Consolidated Bulk Export (pre-existing)

Enqueues a bulk async export for a large set of OOS FSN×DS combinations. Supports 4 query scopes: A=FSN→All DS, B=DS→All FSNs, C=SH→All FSN×DS, D=FSN×SH intersection. Output: B6-schema CSV.

See the overlap panel above — this pre-existing endpoint and the new `consolidated-report` endpoint both do FSN-plus-(DS-or-SH) bulk export. Not consolidated pending a product/backend decision.

Parameters

Parameter	In	Type	Required	Description	Example
queryScope	body	enum	Yes	A=FSN→All DS; B=DS→All FSNs; C=SH→All FSN×DS; D=FSN×SH (AND logic across inputs).	A
fsns	body	string[]	No	Required for scopes A and D.	
darkStores	body	string[]	No	Required for scope B.	
sourcingHubs	body	string[]	No	Required for scopes C and D.	

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/consolidated/export' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Content-Type: application/json' \
--data-raw '{
  "queryScope": "A",
  "fsns": ["ARDET6ZNEMZRRTFK", "LNKXYZ123"]
}'
```

POST /api/v1/diagnostics/pivot — Dynamic Pivot View

This section predates both 2026-07 Figma reads and covers a tab ("Dynamic Pivot View") that neither read captured — see open question 1. Treat everything below as unconfirmed until re-checked.

Returns a dynamic pivot table over the OOS dataset. Accepts a primary pivot dimension (rows), an optional secondary pivot (sub-rows), and metric columns. Used for the Pivot View in Dashboard C.

Parameters

Parameter	In	Type	Required	Description
primaryPivot	body	string	Yes	Row dimension: <code>DARK_STORE</code> <code>FSN</code> <code>BU</code> <code>SC</code> <code>VERTICAL</code> <code>BRAND</code> <code>SOURCING_HUB</code> .
secondaryPivot	body	string	No	Optional sub-row dimension. Same values as <code>primaryPivot</code> .

Parameter	In	Type	Required	Description
metrics	body	string[]	Yes	Columns to include: instockPct oosFsnDsCount scDelayPct cronUnderallocPct inaccurateNormPct stnDropoffPct .
fsnCategory	body	string[]	No	Uber filter.
cityCode	body	string[]	No	City scope filter.
page	body	integer	No	Default: 1.
pageSize	body	integer	No	Default: 50. Max: 200.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/diagnostics/pivot' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Content-Type: application/json' \
--data-raw '{
  "primaryPivot": "DARK_STORE",
  "metrics": ["instockPct", "oosFsnDsCount", "scDelayPct"],
  "cityCode": ["BLR"],
  "fsnCategory": ["T100"]
}'
```

5. Replenishment Execution

15-day live replenishment grid, 3-section cell drill-down modal, STN list, and consignment detail for Dashboard A.

GET /api/v1/replenishment/execution/grid — Live 15-day Grid

Returns the 15-day rolling grid. Source View: rows = Sourcing Hubs. Destination View: rows = Dark Stores. Each cell shows STN count, quantity, and colour code (RED/YELLOW/GREEN/NEUTRAL). Row expansion adds 3 sub-rows: STNs Created, Open, Breaching.

Parameters

Parameter	In	Type	Required	Description	Example
view	query	enum	Yes	SOURCE DESTINATION . Source = SH rows; Destination = DS rows.	SOURCE
city	query	string[]	No	Narrow rows to nodes in these cities.	
sourcingHub	query	string[]	No	Source View only: narrow to these SH codes.	
destinationFc	query	string[]	No	Destination View only: narrow to these DS codes.	
fsnCategory	query	string[]	No	Uber filter.	

Curl

```
curl --location 'http://{instockctl-host}/api/v1/replenishment/execution/grid?view=SOURCE&city=BLR' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/replenishment/execution/cell-summary — Cell Drill-down

Returns the 3-section cell drill-down modal: (1) Execution Status waterfall, (2) DBD Adherence cards, (3) TAT Adherence cards + aging grid. Triggered by clicking any cell in the replenishment grid.

Parameters

Parameter	In	Type	Required	Description	Example
nodeCode	query	string	Yes	SH or DS FC code.	BLR_SH_01
date	query	ISO date "aggregate"	Yes	Calendar day or aggregate across date range.	2026-06-22
view	query	enum	Yes	SOURCE DESTINATION .	SOURCE
rowType	query	enum	Yes	ENTITY CREATED OPEN BREACHING .	ENTITY

TAT aging severity tiers: DEFAULT (0–6h), WARNING (6–24h), CRITICAL (24–36h), EXTREME (>48h). Each clickable card triggers GET /api/v1/replenishment/execution/stns with drillFilter set to the card's category.

Curl


```
curl --location 'http://{instockctl-host}/api/v1/replenishment/execution/cell-summary?nodeCode=BLR_SH_01&date=2026-06-22&view=SOURCE&rowType=ENTITY' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/replenishment/execution/stns — STN List

Paginated, filterable STN list scoped to a cell + drill filter. Sorted: breached first, then by age descending. Each STN row is clickable to open Consignment Detail.

Export CSV: same endpoint with `Accept: text/csv` returns all columns including `exec_status`, `is_dbd_breached`, `dbd_breach_reason`, `is_tat_breached`, `tat_breach_reason`, `tat_breach_aging_bucket`.

Parameters

Parameter	In	Type	Required	Description	Example
nodeCode	query	string	Yes	SH or DS FC code.	
date	query	ISO date "aggregate"	Yes	Calendar day or aggregate.	
view	query	enum	Yes	SOURCE DESTINATION.	
drillFilter	query	string	Yes	execStatus:<stage> dbd:<BREACHED ADHERED YET_TO_ELAPSE> tat:<BREACHED NOT_BREACHED YET_TO_ELAPSE> tatAging:<bucket>.	execStatus:Dispatch Failure
page	query	integer	No	Default: 1.	
pageSize	query	integer	No	Default: 25.	

Curl

```
curl --location 'http://{instockctl-host}/api/v1/replenishment/execution/stns?nodeCode=BLR_SH_01&date=2026-06-22&view=SOURCE&drillFilter=execStatus%3ADispatch%20Failure&page=1&pageSize=25' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/replenishment/execution/stns/{stnId}/consignments — Consignment Detail

Returns header context and all consignments under a selected STN. Opened by clicking an STN row in the STN List modal. Part of the modal stack: Summary → STN List → Consignment Detail. Also reused as the leaf modal from Instock Monitoring's RCA STN rows and from Instock Diagnostics' per-SH STN lists (page 4).

Parameters

Parameter	In	Type	Required	Description	Example
stnId	path	string	Yes	STN identifier.	STN-BLR-20260622-00841

Curl

```
curl --location 'http://{instockctl-host}/api/v1/replenishment/execution/stns/STN-BLR-20260622-00841/consignments' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "stnHeader": {
    "stnId": "STN-BLR-20260622-00841", "fsn": "ARDET6ZNMZRRTFK",
    "sku": "SKU-COLGFRESH200", "sourceFc": "BLR_SH_01", "destinationFc": "BLR_DS_017",
    "stnStatus": "DISPATCHED", "execStatus": "Dispatch Failure",
    "ageHrs": 8.25, "designLtHrs": 8.0, "isBreached": true
  },
  "consignments": [
    {
      "consignmentId": "CONS-BLR-20260622-04412",
      "consignmentStatus": "DISPATCH_FAILED",
      "consignmentQty": 50,
      "stage": "SH Outbound Scan",
      "consignmentCreatedAt": "2026-06-22T06:15:00+05:30",
      "consignmentDispatchedAt": null,
      "boxId": null,
      "consignmentInwardedAt": null,
      "consignmentClosedAt": null
    }
  ]
}
```

Confirmed 2026-07 — `consignments[].stage` **added**. Was previously flagged as an unconfirmed open question (single Figma source). A second, independent Figma read (the Instock Diagnostics "Product View FSN x DS" build prompt) shows the same column again with concrete sample values — "SH Outbound Scan", "Line Haul" — as a pipeline-location string distinct from `consignmentStatus`. Confirm the full enum/value set with backend before UI hardcodes any specific stage names; only the field's existence is confirmed, not its full value list.

6. Cron Monitoring — Summarised View

Always-on allocation health summary for every city, sourcing hub, and network component (Story A1). Driven by Dashboard A uber filters: FSN Category, FSN Portfolio, Store Age, Store OPD, and Date Range (default last 14 days, max 14-day window).

GET /api/v1/cron/summary/cities — City Allocation Summary

Returns one row per city with aggregate allocation quantities, manual IWIT, allocation ratios, a trend indicator, and an inline sparkline series for the active date range. Default sort: `allocPctVsInventoryAwareDeficit` ascending (worst first).

Parameters

Parameter	In	Type	Required	Description	Example
<code>dateFrom</code>	query	ISO date	Yes	Date range start. Max window: 14 days.	<code>2026-06-09</code>
<code>dateTo</code>	query	ISO date	Yes	Date range end (inclusive).	<code>2026-06-22</code>
<code>fsnCategory</code>	query	string[]	No	Uber filter.	<code>T100,T500</code>
<code>fsnPortfolio</code>	query	string[]	No	Uber filter.	
<code>storeAge</code>	query	string[]	No	Uber filter.	
<code>storeOpd</code>	query	string[]	No	Uber filter.	

Key Response Fields

Field	Type	Description
<code>cities[].sumInventoryAwareDeficit</code>	integer	Total <code>min(deficit, available_inventory_at_SH)</code> . Shown in orange — represents fulfillable deficit.
<code>cities[].allocPctVsInventoryAwareDeficit</code>	number null	$(\text{sumAllocatedQty} / \text{sumInventoryAwareDeficit}) \times 100$. Null when <code>sumInventoryAwareDeficit=0</code> .
<code>cities[].trendIndicator</code>	enum	<code>UP</code> =green arrow (improving); <code>DOWN</code> =red arrow (worsening); <code>FLAT</code> =grey dash.
<code>cities[].sparkline[]</code>	Array	One entry per calendar day in the active date range with <code>allocatedQty</code> , <code>deficit</code> , <code>inventoryAwareDeficit</code> , <code>manualIwit</code> . Null fields = no run data (gap).

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/summary/cities?dateFrom=2026-06-09&dateTo=2026-06-22&fsnCategory=T100' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/cron/summary/sourcing-hubs — SH Allocation Summary

One row per sourcing hub with the same allocation metrics as the city view. Includes an SH-exclusive City filter to narrow the list.

Parameters

Parameter	In	Type	Required	Description
dateFrom	query	ISO date	Yes	Date range start.
dateTo	query	ISO date	Yes	Date range end.
city	query	string[]	No	Narrows to SHs in selected cities. Applies only to this view.
fsnCategory	query	string[]	No	Uber filter.

SHs with no run data in date range return 0s for quantity fields and `null` for ratio fields and `trendIndicator` .

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/summary/sourcing-hubs?dateFrom=2026-06-09&dateTo=2026-06-22&city=BLR' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/cron/summary/network-components — Network Component Summary

One row per network component. Includes three NC-exclusive filters: City, Sourcing Hub, Cron Schedule (all multi-select, AND logic, exclusive to this view).

Parameters

Parameter	In	Type	Required	Description	Example
dateFrom	query	ISO date	Yes		
dateTo	query	ISO date	Yes		
city	query	string[]	No	Narrows to NCs in selected cities.	BLR
sourcingHub	query	string[]	No	Narrows to NCs whose SHP0/P1/P2 belongs to selected SH(s).	BLR_SH_01
cronSchedule	query	string[]	No	Narrows to NCs whose CronP0/P1/P2 matches.	Morning
fsnCategory	query	string[]	No	Uber filter.	

Key Response Fields

Field	Type	Description
networkComponents[].ncId	string	Composite NC key, e.g. BLR_K0LAR_Morning_H0SUR_Evening .
networkComponents[].p0Sh	string	Primary sourcing hub code.

Field	Type	Description
networkComponents[].p1Sh	string null	Secondary SH (null if P0-only NC).
networkComponents[].p2Sh	string null	Tertiary SH (null if not present).

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/summary/network-components?dateFrom=2026-06-09&dateTo=2026-06-22&city=BLR&sourcingHub=BLR_SH_01' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

7. Cron Monitoring — Pivoted Deep Dive

Pivot-and-filter workflow (Stories A2–A4). User selects cross-pivot filters, clicks Apply, then gets: a single aggregated Trend View + Constraint Summary (A3), and an 11-tab Run Level View (A4). All endpoints in this section share common pivot filter parameters.

Common Pivot Filter Parameters (all endpoints in this section)

Parameter	In	Type	Required	Description
dateFrom	query	ISO date	Yes	Date range start.
dateTo	query	ISO date	Yes	Date range end.
city	query	string[]	No	City pivot filter.
bu	query	string[]	No	BU filter.
sc	query	string[]	No	Super Category filter.
vertical	query	string[]	No	Vertical filter.
brand	query	string[]	No	Brand filter.
darkStore	query	string[]	No	Dark Store filter.
fsn	query	string[]	No	FSN filter.
sourcingHub	query	string[]	No	Sourcing Hub filter.
networkComponent	query	string[]	No	Network Component filter.
fsnCategory	query	string[]	No	FSN Category uber filter.
fsnPortfolio	query	string[]	No	FSN Portfolio uber filter.
storeAge	query	string[]	No	Store Age uber filter.
storeOpd	query	string[]	No	Store OPD uber filter.

All pivot dimensions are combined with AND logic across dimensions; OR within a single dimension.

GET /api/v1/cron/trend — Replenishment Trend View (Story A3)

Returns the single aggregated trend chart data for the full cross-pivot filter intersection, plus the collapsible aggregated Constraint Summary table. `constraintSummary` is sorted by `constrainedQty` descending; zero-quantity constraints are omitted.

Parameters

All common pivot filter parameters above (dateFrom, dateTo, plus any combination of 11 pivot dimensions).

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/trend?dateFrom=2026-06-09&dateTo=2026-06-22&city=BLR&sc=0ral%20Care' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "trendPoints": [
    {
      "date": "2026-06-22",
      "allocatedQty": 12400, "deficit": 18200,
      "inventoryAwareDeficit": 14800, "inventoryTarget": 62000,
      "manualIwitCreation": 600, "totalRequirementQty": 13000
    }
  ],
  "constraintSummary": [
    {
      "constraint": "INVENTORY_AT_SRC",
      "constrainedQty": 48200,
      "correctiveAction": "Buy inventory at Sourcing Hub."
    }
  ]
}
```

GET /api/v1/cron/run-level — Run Level View / Pivot Tabs (Story A4)

Returns the list of distinct pivot values in scope for the chosen tab. One row per pivot value (no aggregated metrics — rows are clickable entry points to load run lists). All 11 tabs are always rendered.

Additional Parameters

Parameter	In	Type	Required	Description	Example
<code>pivotTab</code>	query	enum	Yes	<code>CITY</code> <code>DARK_STORE</code> <code>BU</code> <code>SC</code> <code>VERTICAL</code> <code>BRAND</code> <code>SOURCING_HUB</code> <code>NETWORK_COMPONENT</code> <code>FSN_CATEGORY</code> <code>FSN_PORTFOLIO</code> <code>FSN</code> .	<code>SC</code>
<code>page</code>	query	integer	No	Default: 1.	
<code>pageSize</code>	query	integer	No	Default: 50.	

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/run-level?dateFrom=2026-06-09&dateTo=2026-06-22&city=BLR&pivotTab=SC' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/cron/run-level/runs — Drill into Runs for a Pivot Value

Returns all individual runs intersecting the selected pivot value x every active pivot filter, sorted by execution date + time descending. Triggered by clicking a pivot value row.

Export CSV (when `exportCsv=true`): includes `run_level_constraint_list` JSON column.

Additional Parameters

Parameter	In	Type	Required	Description	Example
<code>pivotTab</code>	query	enum	Yes	Active pivot tab.	<code>SC</code>
<code>pivotValue</code>	query	string	Yes	The specific pivot value expanded by the user.	<code>Oral Care</code>
<code>page</code>	query	integer	No	Default: 1.	
<code>exportCsv</code>	query	boolean	No	Returns CSV of all visible run rows (ignores pagination).	

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/run-level/runs?dateFrom=2026-06-09&dateTo=2026-06-22&city=BLR&pivotTab=SC&pivotValue=Oral%20Care&page=1' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/cron/run-level/runs/{runId}/constraints — Run-Level Constraint Summary

Returns all constraints that fired for a specific run × pivot value, sourced from the Waterfall Report. Triggered by expanding a run row inside the Run List. Sorted by `constrainedQty` descending. Empty array when no constraint records exist.

Parameters

Parameter	In	Type	Required	Description	Example
<code>runId</code>	path	string	Yes	Run identifier.	<code>RUN-20260622-P2-BLR-001</code>
<code>pivotTab</code>	query	enum	Yes	Active pivot tab.	<code>SC</code>
<code>pivotValue</code>	query	string	Yes	The pivot value of the expanded run row.	<code>Oral Care</code>

Curl

```
curl --location 'http://{instockctl-host}/api/v1/cron/run-level/runs/RUN-20260622-P2-BLR-001/constraints?pivotTab=SC&pivotValue=Oral%20Care' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "runId": "RUN-20260622-P2-BLR-001",
  "constraints": [
    {
      "constraint": "INVENTORY_AT_SRC",
      "constrainedQty": 4800,
      "correctiveAction": "Buy inventory at Sourcing Hub."
    },
    {
      "constraint": "CASE_SIZE",
      "constrainedQty": 1200,
      "correctiveAction": "Deficit present and partially fulfilled. Check if norm/case size needs to be revised."
    }
  ]
}
```


8. Capacity Utilisation

Slot-wise IWIT capacity configuration and consumption across Sourcing Hubs, Dark Stores, and lanes. 30-day rolling window (last 15 to next 15 days). No uber filters apply to this section. Stories A5–A8.

Data sources: `capacityConfigured` comes from MDM (user-uploaded master data, updated manually by ops). `capacityConsumed` and utilisation % are derived from STN execution data written by the pipeline (4h cadence).

GET /api/v1/capacity/sh — SH Capacity View (Story A5)

Returns all sourcing hubs as rows with slot x capacity pivot columns spanning the 30-day window. Each cell contains configured capacity, consumed capacity, and utilisation %. Colour codes: Green=healthy, Amber=under/approaching over, Red=over or not configured.

Parameters

Parameter	In	Type	Required	Description	Example
<code>city</code>	query	string[]	No	Narrow to SHs in selected cities.	<code>BLR</code>
<code>capacityPivot</code>	query	string[]	No	<code>Dry Frozen Chiller Ambient</code> . Default: all.	<code>Dry, Frozen</code>
<code>exportCsv</code>	query	boolean	No	Returns CSV of all visible rows.	

Export CSV columns: `sourcing_hub`, `sh_name`, `city`, `slot_date`, `slot_start`, `slot_end`, `capacity_pivot`, `capacity_configured`, `capacity_consumed`, `pct_capacity_utilised`, `missing_capacity_reason`.

`missingCapacityReason` values: `OB_CAPS_MISSING` | `IB_CAPS_MISSING` | `STORAGE_NOT_CONFIGURED` | `TRANSPORT_LANE_NOT_PRESENT`.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/capacity/sh?city=BLR&capacityPivot=Dry' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/capacity/ds — DS Capacity View (Story A6)

Returns all dark stores as rows with the same slot x capacity pivot column structure. Adds a Connected Sourcing Hub filter.

Parameters

Parameter	In	Type	Required	Description
<code>city</code>	query	string[]	No	Narrow to DSs in selected cities.
<code>connectedSourcingHub</code>	query	string[]	No	Narrow to DSs connected to selected SH(s).
<code>capacityPivot</code>	query	string[]	No	<code>Dry Frozen Chiller Ambient</code> . Default: all.
<code>exportCsv</code>	query	boolean	No	

Curl

```
curl --location 'http://{instockctl-host}/api/v1/capacity/ds?connectedSourcingHub=BLR_SH_01&capacityPivot=Dry' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/capacity/lane — Lane Level View (Story A7)

Returns capacity and utilisation broken down by SH → DS transport lane for a given slot. Shows imbalances in lane-level capacity allocation.

Parameters

Parameter	In	Type	Required	Description	Example
sourcingHub	query	string	Yes	SH FC code.	BLR_SH_01
slotDate	query	ISO date	Yes	Slot date.	2026-06-22
slotStart	query	string	Yes	Slot start (HH:mm).	08:00
slotEnd	query	string	Yes	Slot end (HH:mm).	14:00
capacityPivot	query	string	Yes	Capacity pivot.	Dry
exportCsv	query	boolean	No		

Curl

```
curl --location 'http://{instockctl-host}/api/v1/capacity/lane?sourcingHub=BLR_SH_01&slotDate=2026-06-22&slotStart=08:00&slotEnd=14:00&capacityPivot=Dry' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/capacity/cell-drilldown — Capacity Cell Drill-Down (Story A8)

Returns summary KPIs and the top-level capacity breakdown for a clicked cell — consumed vs configured, plus contribution from each utilisation category.

Parameters

Parameter	In	Type	Required	Description	Example
nodeCode	query	string	Yes	SH or DS FC code.	BLR_SH_01
slotDate	query	ISO date	Yes		2026-06-22
slotStart	query	string	Yes	Slot start (HH:mm).	08:00
slotEnd	query	string	Yes	Slot end (HH:mm).	14:00
capacityPivot	query	string	Yes	Capacity pivot.	Dry

Curl

```
curl --location 'http://{instockctl-host}/api/v1/capacity/cell-drilldown?nodeCode=BLR_SH_01&slotDate=2026-06-22&slotStart=08:00&slotEnd=14:00&capacityPivot=Dry' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/capacity/cell-drilldown/under-utilisation — Section 2a: Cron Under-Utilisation

Returns the cron under-utilisation breakdown waterfall for a capacity cell — showing how much allocated capacity was not fulfilled and why.

Parameters

Same as cell-drilldown: `nodeCode`, `slotDate`, `slotStart`, `slotEnd`, `capacityPivot`, plus optional `exportCsv`.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/capacity/cell-drilldown/under-utilisation?nodeCode=BLR_SH_01&slotDate=2026-06-22&slotStart=08:00&slotEnd=14:00&capacityPivot=Dry' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

GET /api/v1/capacity/cell-drilldown/wasted — Section 2b: Capacity Reserved but Not Usable

Returns a waterfall of wasted capacity reasons: Seller Not Onboarded in WMS, STN Cancellation Failed, STN Dispatched but Returned / Rejected, Other. Per-STN attribution is included inline; no separate endpoint required.

Parameters

Same as cell-drilldown: `nodeCode`, `slotDate`, `slotStart`, `slotEnd`, `capacityPivot`, plus optional `exportCsv`.

Export CSV columns: `node_code`, `slot_date`, `slot_start`, `slot_end`, `capacity_pivot`, `reason`, `stn_id`, `fsn`, `source_fc`, `destination_fc`, `sku`, `seller_id`, `quantity`, `stn_status`, `stn_created_at`, `dbd`.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/capacity/cell-drilldown/wasted?nodeCode=BLR_SH_01&slotDate=2026-06-22&slotStart=08:30&slotEnd=14:30&capacityPivot=Dry' \
--header 'Authorization: Bearer {sso-token}' \
--header 'Accept: application/json'
```

Example Response

```
{
  "waterfall": [
    {
      "reason": "Seller Not Onboarded in WMS",
      "capacityWastedUnits": 120,
      "stns": [
        {
          "stnId": "STN-BLR-20260622-00912", "fsn": "LNKXYZ123",
          "sourceFc": "BLR_SH_01", "destinationFc": "BLR_DS_022",
          "sku": "SKU-GILLSHIELD100", "sellerId": "FK-SELLER-042",
          "quantity": 120, "stnStatus": "CANCELLED",
          "stnCreatedAt": "2026-06-22T07:00:00+05:30",
          "dbd": "2026-06-22T12:00:00+05:30"
        }
      ]
    },
    { "reason": "STN Cancellation Failed", "capacityWastedUnits": 0, "stns": [] },
    { "reason": "STN Dispatched but Returned / Rejected", "capacityWastedUnits": 0, "stns": [] },
    { "reason": "Other", "capacityWastedUnits": 0, "stns": [] }
  ],
}
```

```
"totalWasted": 120
```

```
}
```

9. Bulk Actions

Bulk action endpoint for applying approve / reject / override decisions on replenishment requirements in batch. Wraps the internal ip-return bulk action API. Accepts a CSV file upload via multipart form data.

Upstream reference: POST `http://10.24.36.138/ip-return/v1/requirement/bulk_action` (production ip-return service). Meridian proxies this to keep auth and audit in one place.

POST /api/v1/replenishment/bulk-action — Bulk Approve / Reject / Override Requirements

Accepts a CSV file of requirement identifiers plus an `actionType` field. Processes all rows asynchronously and returns a job summary. Intended for use from the Cron Monitoring and Replenishment Execution views when an operator wants to act on multiple STN requirements at once.

Parameters

Parameter	In	Type	Required	Description	Example
<code>actionType</code>	form-data	string	Yes	<code>APPROVE REJECT OVERRIDE</code> .	<code>APPROVE</code>
<code>file</code>	form-data	file (CSV)	Yes	CSV with columns: <code>requirement_id</code> , <code>fsn</code> , <code>dark_store</code> , <code>sourcing_hub</code> , <code>quantity</code> . Max 5,000 rows per request.	
<code>overrideValue</code>	form-data	number	No	Required when <code>actionType=OVERRIDE</code> . New quantity to set.	<code>150</code>
<code>reason</code>	form-data	string	No	Free-text reason for the action. Stored in audit log.	<code>Manual override – festive surge</code>

CSV Column Schema

Column	Type	Required	Description
<code>requirement_id</code>	string	Yes	TR / STR identifier from the replenishment system.
<code>fsn</code>	string	Yes	FSN identifier.
<code>dark_store</code>	string	Yes	Destination DS FC code.
<code>sourcing_hub</code>	string	Yes	Source SH FC code.
<code>quantity</code>	integer	Yes	Current allocated quantity (used for validation).

Response Schema

Field	Type	Description
<code>jobId</code>	string (UUID)	Async job ID. Poll <code>GET /api/v1/exports/{jobId}</code> for completion status.

Field	Type	Description
status	QUEUED	Always QUEUED on submission.
totalRows	integer	Total CSV rows parsed.
validRows	integer	Rows that passed validation and were queued.
invalidRows	integer	Rows rejected due to validation errors (unknown ID, wrong quantity, etc.).
validationErrors	Array<{row, reason}>	First 20 validation errors for immediate feedback.
estimatedSeconds	integer	Estimated processing time.

Curl

```
curl --location 'http://{instockctl-host}/api/v1/replenishment/bulk-action' \
--header 'Authorization: Bearer {sso-token}' \
--form 'actionType="APPROVE"' \
--form 'reason="Confirmed via ops review"' \
--form 'file=@"/path/to/requirements.csv"'
```

Example Response (2xx)

```
{
  "jobId": "d4e5f6a7-b8c9-0123-def0-456789abcdef",
  "status": "QUEUED",
  "totalRows": 120,
  "validRows": 118,
  "invalidRows": 2,
  "validationErrors": [
    { "row": 34, "reason": "requirement_id 'TR-2026-BLR-UNKNOWN' not found" },
    { "row": 87, "reason": "quantity mismatch: supplied 50, current value is 75" }
  ],
  "estimatedSeconds": 30
}
```

Example Response (4xx — file validation failure)

```
{
  "type": "urn:meridian:error:bad-request",
  "title": "Bad Request",
  "status": 400,
  "detail": "CSV file exceeds maximum row limit of 5000.",
  "instance": "/api/v1/replenishment/bulk-action"
}
```

Navigation Flows & Frontend Cheat Sheet

Every click-chain in the app mapped to its API call sequence. Use this for your design team presentation and as the reference for frontend engineers.

How Uber Filters Work

Uber filters are NOT separate API calls. They are query parameters passed along with every API call.

When the user selects `FSN Category = T100, T500` in the filter bar, the frontend stores `fsnCategory=T100,T500` in URL/state and every subsequent call includes those params. The backend filters server-side.

User selects: FSN Category = T100 + Store Age = >60

Then:

```
GET /instock/kpi?fsnCategory=T100&storeAge=>60          ← KPI cards filtered
GET /instock/cities?fsnCategory=T100&storeAge=>60       ← city table filtered
GET /instock/cities/BLR/sub-pivots?...&fsnCategory=T100&storeAge=>60
GET /instock/rca/supply-chain?...&fsnCategory=T100&storeAge=>60
```

Every call carries the same filter params. No separate "apply filter" API.

When the user changes a filter: frontend re-fetches everything currently visible in parallel. Each re-fetch includes the updated params.

Scope of uber filters: ALL Dashboard B endpoints accept uber filters. Dashboard C and Dashboard A do NOT use them — they have their own filter mechanisms.

Page Load — What Fires on First Open

Dashboard B (Instock Monitoring) — Page Load

```
User opens Dashboard B
├─ GET /api/v1/status
│   Returns: lastRunTimestamp, isHeld, holdReason
│   Frontend shows: "Data as of 2:30 PM" + hold banner if isHeld=true
├─ GET /api/v1/filters/options
│   ?sections=fsnCategories,fsnPortfolios,storeAgeBands,storeOpdBands
│   Populates: the 4 uber filter dropdowns
│   Called ONCE on page load. Not called again unless page is refreshed.
├─ GET /api/v1/instock/kpi
│   Returns: 3 KPI cards with 4 deltas each
└─ GET /api/v1/instock/cities
    Returns: city table rows (one per city)

Total on page load: 4 parallel API calls
```

Flow 1: KPI Cards Drill-Down

"View Worst FSNs" button (on OOS Count card)

```
GET /api/v1/instock/kpi/worst-fsns
├─ Returns: FSN rows ranked by OOS breadth %
├─ Columns: FSN, product name, OOS store count, total stores, breadth %, top cause
└─
```

```
↳ user clicks any FSN row
  ↳ GET /api/v1/instock/product-drilldown
    ?scope=FSN&fsn=ARDET6ZNEMZRRTFK
    Returns: all OOS FSN×DS pairs for that FSN
    (from here, user can click bucket % → RCA modals, see Flow 3)
```

"View List" button (on DS Below 90% card)

```
GET /api/v1/instock/kpi/low-instock-stores
| Returns: dark store rows sorted by instock % ascending
| Columns: DS code, DS name, city, instock %, OOS count, top cause
|
↳ user clicks any DS row
  ↳ GET /api/v1/instock/product-drilldown
    ?scope=DARK_STORE&darkStoreCode=BLR_DS_017
    Returns: all OOS FSN×DS pairs at that store
```

Flow 2: City Table → Geography Drill-Down

```
GET /api/v1/instock/cities
| Returns: one row per city
|
| — Path A: User selects View Filter (e.g. "Vertical")
|   ↳ user clicks expand arrow on "Bengaluru" row
|     ↳ GET /instock/cities/BLR/sub-pivots
|       ?pivotType=VERTICAL
|       Returns: nested rows (Grocery, Electronics, etc.) under BLR
|         |
|         ↳ user clicks expand on "Grocery" row
|           ↳ GET /instock/cities/BLR/dark-stores
|             ?pivotType=VERTICAL&pivotValue=Grocery
|             Returns: DS rows within BLR × Grocery
|               |
|               ↳ user clicks "Product Drilldown" button on a DS row
|                 ↳ GET /instock/product-drilldown
|                   ?scope=DARK_STORE&darkStoreCode=BLR_DS_001
|                   &pivotType=VERTICAL&pivotValue=Grocery
|
| — Path B: User clicks "Product Drilldown" directly on city row (no View Filter)
|   ↳ GET /instock/product-drilldown
|     ?scope=CITY&cityCode=BLR
|     Returns: all OOS FSN×DS in Bengaluru
|
| — Path C: User clicks any bucket % → opens RCA modal (see Flow 3 below)
```

The `scope` param tells the API WHERE you came from:

- `scope=CITY&cityCode=BLR` → all OOS in Bengaluru
- `scope=SUB_PIVOT&cityCode=BLR&pivotType=VERTICAL&pivotValue=Grocery` → all OOS in BLR × Grocery
- `scope=DARK_STORE&darkStoreCode=BLR_DS_001` → all OOS at this one store

This same `scope` pattern is used by ALL RCA modals — they inherit the scope from wherever you clicked.

Flow 3: The 4 RCA Modals (B7.1 — B7.4)

These open when you click any non-zero bucket % at ANY level (city row, sub-pivot row, DS row, or product drilldown FSN row). The `scope` params carry through from where you clicked.

B7.1 — Supply Chain Delay (4 APIs, deepest chain)

```
User clicks "SC Delay 42.1%" on Bengaluru row
|
```


↳ GET /instock/rca/supply-chain
?scope=CITY&cityCode=BLR
Returns: SC labels ranked table (10 stages)

SC Issue	# FSN×DS	Impacted Qty	% SC OOS
TL not created	14	252	18.2%
Inv On Hold (Stock take,GTL)	10	110	13.0%

↳ user clicks "TL not created" row (EXPAND)
↳ SAME API + expandIssue param
?scope=CITY&cityCode=BLR&expandIssue=TL%20not%20created
Returns: STN rows nested under that label

STN ID	FSN	Source FC	Dest FC	Qty	Status
STN-9810	F000068	BOM_SH_BHI	BOM_DS_026	57	At DS

"Click any row to view consignment list"

↳ user clicks ANY COLUMN in STN row (entire row is clickable)
↳ GET /replenishment/execution/stns/{stnId}/consignments
Returns: consignment detail
Shows: "← Back to STN list" breadcrumb (frontend nav, no API)

Consignment ID	Consignment Status	Qty	Created At
CON-8023220	Dispatched	13	2026-06-25
CON-9245601	Inwarded	134	2026-06-23

EXPORT: "Export All SC Issues (CSV)" button at bottom
↳ POST /exports { exportType: "SC_RCA", filters: {scope params} }

FILTERS WITHIN MODAL: Sourcing Hub dropdown + Dark Store dropdown
↳ same GET /rca/supply-chain + &sourcingHub=X&darkStore=Y
(re-fetches the modal data filtered by SH/DS)

Frontend note: The entire STN row is clickable (not just STN ID). "← Back to STN list" is frontend navigation only — no API call.

B7.2 — Cron Underallocation (3 APIs + frontend navigation)

User clicks "Cron Underalloc 28.5%" on Bengaluru row

↳ GET /instock/rca/cron-underallocation
?scope=CITY&cityCode=BLR
Returns: 19 constraint labels ranked table

Cron Issue	# FSN×DS	Design Norm	Impacted Qty
Min Ideal Quantity	9	12 units	
NO_DEFICIT_CASESIZE	8	24 units	64

↳ user clicks "Min Ideal Quantity" row (EXPAND)
↳ SAME API + expandIssue param
?scope=CITY&cityCode=BLR&expandIssue=Min%20Ideal%20Quantity
Returns: FSN rows nested under that constraint

FSN	Product	DS	Qty	P0 SH	P0 Cron Phase
F000180	Lactogen	DS-027	62	BOM_VASAI	2026-05-06

"Click any FSN to open Dashboard C diagnostic"

↳ user clicks BLUE FSN link (F000180XYZ)
↳ FRONTEND NAVIGATES to Dashboard C (new page, modal closes)
URL: /diagnostics/product-view?fsn=F000180XYZ&darkStoreCode=DS-027
↳ GET /diagnostics/fsn-ds
?fsn=F000180XYZ&darkStoreCode=DS-027
(this call happens on the new page, not from the modal)

EXPORT: "Export All Cron Issues (CSV)" button
↳ POST /exports { exportType: "CRON_RCA" }

Frontend note: FSN is the ONLY clickable column (others are read-only). Clicking FSN navigates AWAY from the modal to Dashboard C. The FSN expansion response includes a `darkStore` field — use it to pre-fill the Dashboard C URL.

B7.3 — Inaccurate Norm / Demand MAPE (3 APIs + frontend navigation)

User clicks "Inaccurate Norm 18.3%" on Bengaluru row

- ↳ GET `/instock/rca/inaccurate-norm`
`?scope=CITY&cityCode=BLR`
Returns: 5 MAPE bands ranked table

MAPE Band	# FSN×DS	Design Norm	Impacted
100–125%	12	varies	
125–150%	8	varies	
150–175%	5	varies	

- ↳ user clicks "150–175" band (EXPAND)
 - ↳ SAME API + `expandBand` param
`?scope=CITY&cityCode=BLR&expandBand=150–175`
Returns: FSN rows with norm vs actual sales
 - ↳ user clicks BLUE FSN link
 - ↳ FRONTEND NAVIGATES to Dashboard C (same as B7.2)
 - ↳ GET `/diagnostics/fsn-ds`

EXPORT: "Export MAPE Report (CSV)" button

- ↳ POST `/exports` { `exportType: "NORM_RCA"` }

Behavior identical to B7.2 — only FSN is clickable, navigates to Dashboard C.

B7.4 — STN Creation Drop-Off (2 APIs, dead end)

User clicks "STN Drop-Off 11.1%" on Bengaluru row

- ↳ GET `/instock/rca/stn-creation-dropoff`
`?scope=CITY&cityCode=BLR`
Returns: 35 failure stages ranked table

STN Creation Drop Off Stage	# FSN×DS	Impacted Qty	% of OOS
Error while fetching from RLM	8	136	9.8%
Product details not found	7	126	8.5%

- ↳ user clicks any stage row (EXPAND)
 - ↳ SAME API + `expandStage` param
`?scope=CITY&cityCode=BLR&expandStage=1`
Returns: TR rows (Transfer Requirement detail)

TR/STR ID	FSN	Source FC	Dest FC	Qty	TR Status
TR-69173	F000032	BOM_THANE	BOM_DS_028	125	Pend Pick

🔴 NOTHING IS CLICKABLE. TR rows are read-only terminal data.

EXPORT: "Export All STN Issues (CSV)" button

- ↳ POST `/exports` { `exportType: "STN_RCA"` }

Frontend note: TR rows are terminal — no click-through. Do NOT add click handlers on these rows.

Flow 4: Dashboard C — Instock Diagnostics

C1: FSN×DS Diagnostic (standalone page)

Capacity Utilisation (A5–A8)

```
User opens Capacity tab
├─ GET /capacity/sh
│   Returns: SH × 30-day grid (rows=SHs, cols=time slots)
├─ GET /capacity/ds
│   Returns: DS × 30-day grid
└─ user clicks a lane
    └─ GET /capacity/lane
        ?direction=SH_VIEW&nodeCode=BLR_SH_01
        Returns: lane-level grid
        └─ user clicks any cell (3 parallel API calls)
            ├─ GET /capacity/cell-drilldown
            │   Returns: header + consumed STN list (Section 1)
            ├─ GET /capacity/cell-drilldown/under-utilisation
            │   Returns: 5-reason waterfall (Section 2a)
            └─ GET /capacity/cell-drilldown/wasted
                Returns: 4-reason waterfall (Section 2b)
```

Replenishment Execution (A9)

```
GET /replenishment/execution/grid
?view=SOURCE (or DESTINATION)
├─ Returns: 15-day grid with color-coded cells
└─ user clicks any cell
    └─ GET /replenishment/execution/cell-summary
        ?nodeCode=BLR_SH_01&date=2026-06-22&view=SOURCE
        Returns: ALL 3 sections in ONE response:
            Section 1: Execution Status (14 stages)
            Section 2: DBD Adherence (5 cards)
            Section 3: TAT Adherence (5 cards + aging buckets)
        └─ user clicks any card in Section 1/2/3
            └─ GET /replenishment/execution/stns
                ?drillFilter=execStatus:Pending%20To%20Pick
                Returns: paginated STN list
                └─ user clicks any STN row
                    └─ GET /replenishment/execution/stns/{stnId}/consignments
                        Returns: consignment detail
                        (same endpoint as B7.1's deepest level – shared)
```

Flow 6: Download Manager (shared)

```
User clicks "↓ Downloads" in sidebar
└─ GET /exports
    Returns: all export jobs (QUEUED/PROCESSING/READY/FAILED/EXPIRED)

Auto-refresh: while any job is QUEUED or PROCESSING
└─ GET /exports/{jobId}
    every 15 seconds until status changes

User clicks "Download" on a READY job
└─ browser opens downloadUrl (pre-signed GCS link, no API call)
```

What's Missing From the API Contract

These behaviors are implied by design but not explicitly documented in the spec. Frontend engineers should note them:

#	What	Where it should go	Why it matters
1	B7.2/B7.3: Only FSN column is clickable (blue link). Navigates to Dashboard C pre-filling <code>fsn + darkStore</code> .	After cron-underallocation and inaccurate-norm response schema	Frontend needs to know which column to make a link
2	B7.4: TR rows are terminal — no click-through behavior.	After stn-creation-dropoff response schema	Frontend should NOT add click handlers
3	B7.1: Entire STN row is clickable, not just STN ID.	After supply-chain RCA response schema	Frontend should add row-level <code>onClick</code> , not column-level
4	"← Back to STN list" is frontend navigation (no API call)	After consignment detail endpoint	Frontend manages modal breadcrumb state internally
5	Uber filters re-fetch all visible components in parallel when changed	Before Dashboard B KPI section	Frontend must track which components are mounted and re-fetch all